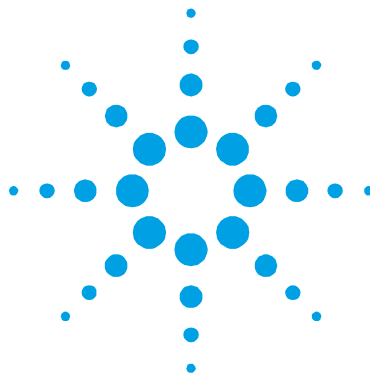




# **Dual Stage Rotary Vane Pumps**

**DS102 1Ph, DS202 1Ph, DS302 1Ph,  
DS402 1Ph, DS402 3Ph, DS602 1Ph,  
DS602 3Ph**

## **Models**

**9499315, 9499320, 9499325, 9499330,  
9499331, 9499335, 9499336**

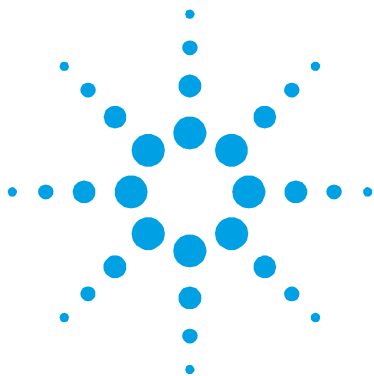
**Manuale di istruzioni  
Bedienungshandbuch  
Notice de mode d'emploi  
User Manual**

**87-900-937-01 (P)**

**08/2015**



**Agilent Technologies**



## 4

# Instructions for Use

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Original Instructions



## **General Information**

This equipment is destined for use by professionals. The user should read this instruction manual and any other additional information supplied by Agilent before operating the equipment. Agilent will not be held responsible for any events occurring due to non-compliance, even partial, with these instructions, improper use by untrained persons, non-authorized interference with the equipment or any action contrary to that provided for by specific national standards.

The DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph and DS602 3Ph pumps are dual-stage, rotary vane pumps oil sealed, driven by a single-phase or three-phase electric motor.

These high vacuum pumps are suitable for pumping non corrosive gases.

The following paragraphs contain all the information necessary to guarantee the safety of the operator when using the equipment. Detailed information is supplied in the appendix "Technical Information".

**This manual uses the following standard protocol:**

---

**WARNING!**



Warning messages are for attracting the attention of the operator to a particular procedure or practice which, if not followed correctly, could lead to serious injury.

---

**CAUTION!**

The caution messages are displayed before procedures which, if not followed, could cause damage to the equipment.

---

**NOTE**

Notes contain important information taken from the text.

---

## Symbols used

The following symbols are used consistently throughout in all illustration:

| Symbols                                                                             | Description                                                      |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
|    | Suction flange                                                   |
|    | Exhaust flange                                                   |
|    | Filling up operating fluid                                       |
|    | Gas Ballast valve                                                |
|    | Caution, hot surface: “Danger of burns if hot parts are touched” |
|   | Warning “see installation/operating mode instruction”            |
|  | Caution, risk of electric shock                                  |

## Storage

When transporting and storing the pumps, the following environmental requirements should not be exceeded:

- temperature: from -20° to +70 °C
- relative humidity: 0 – 95 % (non-condensing)

## Preparation for Installation

The pump is supplied in a special protective packing. If this shows signs of damage which may have occurred during transport, contact your local sales office.

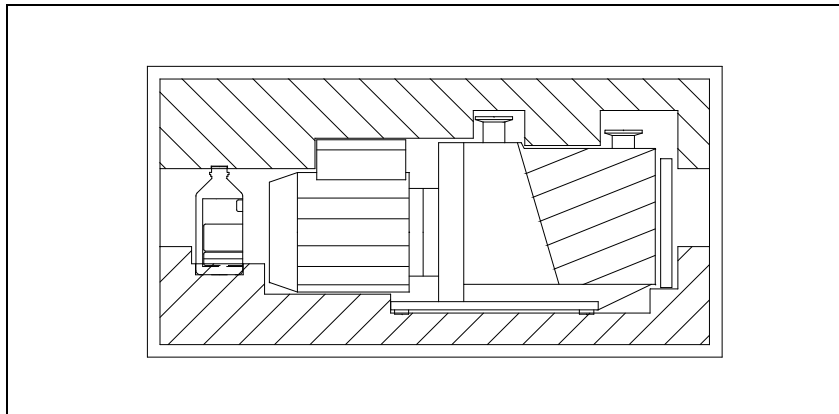
Total weight of the pack, including the pump, is approx. 32 Kg.

When unpacking the pump, be sure not to drop it and avoid any kind of sudden impact or shock vibration to it.

Do not dispose of the packing materials in an unauthorized manner. The material is 100 % recyclable and complies with EEC Directive 85/399.

### NOTE

Normal exposure to the environment cannot damage the pump. Nevertheless, it is advisable to keep it closed until it is installed in the system, thus preventing any form of pollution by dust.



**Figure 1**

## Installation

Do not install or use the pump in an environment exposed to atmospheric agents (rain, snow, ice), dust, aggressive gases, or in explosive environments or those with a high fire risk. During operation, the following environmental conditions must be respected:

- temperature: from +12 °C to +40 °C
- relative humidity: 0 – 95 % (non-condensing)

WARNING!



Protect against short circuits and overload by installing on Agilent Device electrical main line an automatic circuit breaker Type C of proper capacity (see table here below):

| P/N     | 110 Vac | 220 Vac | 380 Vac |
|---------|---------|---------|---------|
| 9499315 | 10 A    | 6 A     |         |
| 9499320 | 10 A    | 6 A     |         |
| 9499325 | 10 A    | 6 A     |         |
| 9499330 | 16 A    | 7.5 A   |         |
| 9499331 |         | 6 A     | 6 A     |
| 9499335 | 16 A    | 7.5 A   |         |
| 9499336 |         | 6 A     | 4 A     |

WARNING!



The pump must be installed in a way that allows an easy interruption of the line voltage.

Power supply cord: The correct cable for electrical wiring is a three wires (Ph+N+Earth) cable. The wire section has to be at least 0.75 mm<sup>2</sup> (AWG18).



## 4 Instructions for Use

### Installation

#### CAUTION!

Before starting the pump, fill up with lubricating oil as the pump is delivered empty.

---

#### WARNING!



Remove protective cap from inlet flange. Remove metallic cap and rubber membrane from exhaust flange following the procedure showed in the photo sequence.

---

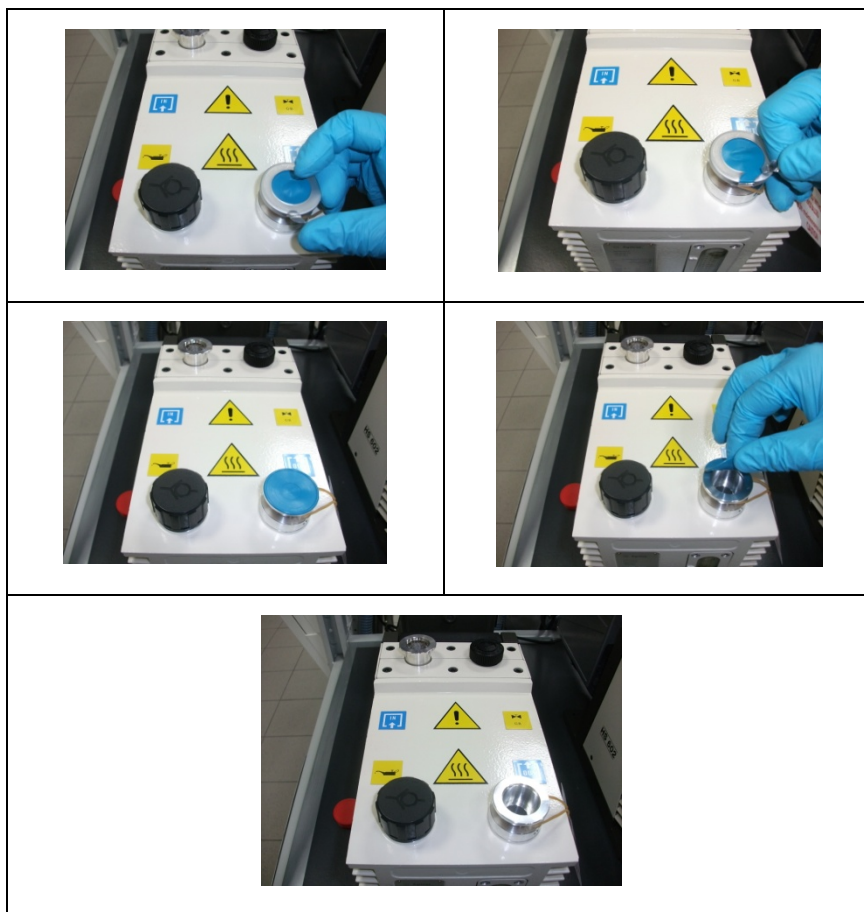


Figure 2

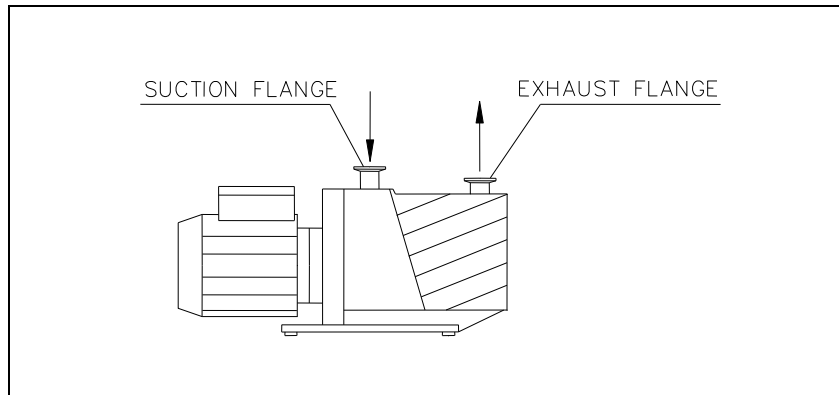
## 4 Instructions for Use

### Installation

#### **WARNING!**



During installation, pay maximum attention that the suction flange is connected to the vacuum chamber and the exhaust flange is not closed (see the following figure). The pump must not be used as a compressor.



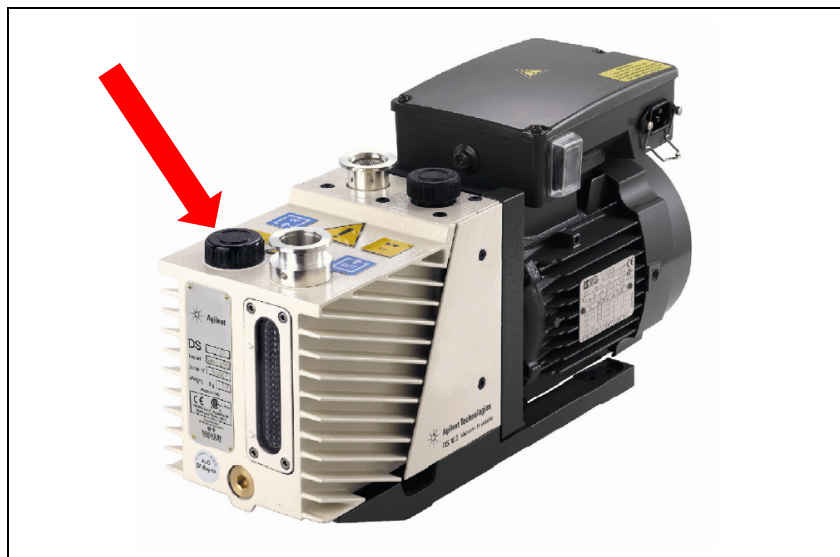
**Figure 3**

**WARNING!**



A safety device (indicated by the red arrow in the figure below) is installed on the pump to guarantee that the pressure inside the oil vessel never exceeds the maximum permitted value of 1.7 bar (abs). This is a safety device and for this reason the cap must not be replaced, damaged or tampered with. Temporary removal (by unscrewing) is allowed to fill oil into the pump when needed. Users are responsible for screwing the device back in its original position after filling and before starting the pump. Failure to respect these precautions may cause damage to the machine and injury to the operator.

A mixture of intake gas and oil will issue from the safety cap if the pressure in the oil vessel exceeds the allowed limit values (e.g. obstructed exhaust). The flow will continue until the pump is stopped and/or the cause of overpressure is removed (e.g. obstruction is removed from exhaust).



**Figure 4**

## 4 Instructions for Use

### Use

#### CAUTION!

Check that your electrical mains voltage corresponds to that indicated on the motor's plate.

---

Connect the pump to the power supply.

#### CAUTION!

Pay special attention to the three-phase motor, where an inversion of polarity causes inversion of the direction of rotation of the pump with consequent possibilities of mechanical damage.

---

## Use

Before being put into service, in order to reach maximum vacuum, the pump must be left running for about an hour with the gas ballast valve open. This will eliminate any humidity from the oil.

There are no special procedures for switching the pump on; it needs only to be connected to the electric power by means of the bipolar switch (version with single-phase motor) or of the multipolar switch (version with three-phase motor).

---

#### WARNING!



**The pump is designed for operation with neutral or non-corrosive fluids. It is absolutely forbidden to use potentially explosive or flammable substances.**

---

There are no special procedures for switching the pump off; it needs only to be disconnected from the electric power by means of the bipolar switch (version with single-phase motor) or of the multipolar switch (version with three-phase motor).

## Maintenance

Personnel responsible for pump operation and maintenance must be well-trained and must be aware of the accident prevention rules.

---

**WARNING!**



Death may result from contact with high voltages. Always take extreme care and observe the accident prevention regulations in force.

---

---

**WARNING!**



When machine is powered take care on account of moving parts and high voltages.

---

---

**WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.

---

## 4 Instructions for Use

### Maintenance

---

#### **WARNING!**



Always disconnect the power supply to the pump before starting maintenance work. Place a special warning signs over the power supply breaker switch: **MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON**. When finished, remove the safety warning.

---

#### **WARNING!**



Do not change the oil immediately after stopping the machine as the oil temperature may still be high.

---

#### **CAUTION!**

When replacing a part, take great care. Pay special attention to the three-phase motor, where an inversion of polarity causes inversion of the direction of rotation of the pump with consequent possibilities of mechanical damage.

---

#### **NOTE**

Before returning the pump to the constructor for repairs the "Health and Safety" sheet attached to this instruction manual must be filled-in and sent to the local sales office. A copy of the sheet must be inserted in the pump package before shipping.

---

If a pump is to be scrapped, it must be disposed of in accordance with the specific national standards.

## Disposal

### Meaning of the "WEEE" logo found in labels

The following symbol is applied in accordance with the EC WEEE (Waste Electrical and Electronic Equipment) Directive.

This symbol (**valid only in countries of the European Community**) indicates that the product it applies to must NOT be disposed of together with ordinary domestic or industrial waste but must be sent to a differentiated waste collection system.

The end user is therefore invited to contact the supplier of the device, whether the Parent Company or a retailer, to initiate the collection and disposal process after checking the contractual terms and conditions of sale.



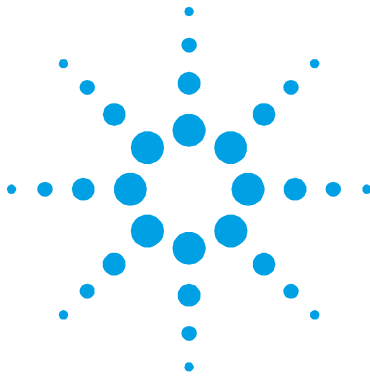
For more information refer to:

<http://www.agilent.com/environment/product/index.shtml>



## **4    Instructions for Use**

### **Disposal**



## 5 Technical Description

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Original Instructions



## **5    Technical Description**

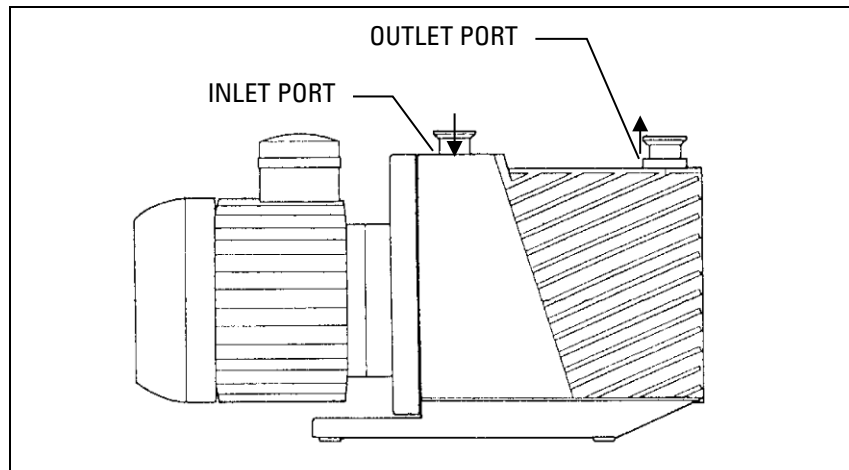
### Disposal

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## SECTION I

### Technical Description

The DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph and DS602 3Ph pumps are dual-stage, rotary vane pumps oil sealed, driven by a single-phase or three-phase electric motor.



**Figure 5**

These vacuum pumps are suitable for pumping non corrosive gases.

The main features are:

- all parts in direct contact with the fluid pumped are free of copper alloys;
- all materials are carefully selected to provide extended life;
- a high capacity gas ballast device allow to pump condensable vapors;
- all the parts composing the pump are fully replaceable due to the close machining tolerances and to the centering obtained by using reference pins;

## 5 Technical Description

### Technical Description

- due to its design features and low number of gaskets, the pump requires little maintenance, disassembly and reassemble are easy and require minimal time.

The pump works with force-feed lubrication, provided by an auxiliary gear-pump driven by the rotor of the vacuum pump itself. This ensures proper lubrication even when pressures are close to atmospheric.

The entire pump functional block is immersed in the oil contained in the casing. The oil guarantees perfect sealing of the discharge valves, enters the pump to ensure lubrication and sealing of the parts inside, facilitates heat dissipation and reduces pump noise.

The pump is equipped with a special anti-suckback device which automatically isolates the vacuum system when the pump stops. This avoids rises in pressure or oil flow in the vacuum system while air is allowed back into the stator chambers.

The air entering the pump after the anti-suckback device has closed prevents the oil in the casing from filling the stator chambers.

The inner seals are achieved by the lubricating oil, thanks to the close machining tolerances.

There is only one oil seal on the rotor shaft, the one preventing oil from seeping out of the pump. This seal acts on a bushing fitted onto the shaft. Simple replacement of this seal grants the rotor a practically unlimited life (pumps DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph). On the DS102 1Ph, DS202 1Ph and DS302 1Ph pumps, this seal acts on a surface-hardened area of the shaft guaranteeing long rotor life.

The vacuum pumps are connected directly to the electric motor through a flexible coupling, so that motion is transmitted even in case of poor alignment.

A fan fitted on the joint and housed in the pump support produces a forced air flow over the finned surfaces of the casing to avoid oil overheating.

## Lubrication

The pump's lubricating system is force-feed type.

Oil circulation is obtained by means of a gear pump connected to the rotor shaft.

The oil is drawn through a gauze filter to prevent any foreign bodies from entering the pump.

Shunt-mounted on the delivery line is a hydraulic piston which, besides actuating the anti-suckback device (see ANTI-SUCKBACK DEVICE), also regulates pressure by discharging excess oil flow directly into the tank.

The required flow of oil under pressure passes through ducts that are drilled in the walls of the pump and lubricates the bushings and the inside parts.

## Vacuum Seals

A special feature of this pump is the low number of gaskets that are employed.

The seals in the circuit are obtained by means of VITON gaskets.

The careful surface finish of the various parts of the pump means that vacuum sealing of the functional block is ensured by the film of oil separating metal surfaces. Sealing of the rotor shaft is guaranteed by a rotating gasket with dust-guard lip.

The suction flange and duct are sealed by mean of OR gaskets.

## Gas Ballast Valve

When the pump sucks in vapors, these condense during compression and mix with the oil, forming an emulsion (an aqueous vapor, for example) or a solution (organic solvent vapors, for example).

A number of problems arise from this, such as the impossibility of obtaining high vacuums, the alteration of the properties of the oil, could cause scaling on parts of the pump.

To avoid this, during compression at the second stage, atmospheric air is let into the pump through an adjustable valve, or “gas ballast valve”, located at the top of the pump.

In this way, the discharge valve of the second stage opens through the effect of the atmospheric air before the partial pressure of the vapor reaches saturation point, thus preventing condensation from occurring. The vapors are expelled mixed with air.

## **Anti-Suckback Device**

The pump is equipped with a special anti-suckback device to avoid air pressure rises and/or oil backflow towards the evacuated vessel when the pump is switched off. This device has a shutter which automatically closes the suction duct.

In this way the pump and vacuum system are completely isolated from each other and air can enter the pump without any risk for the vacuum produced in the system.

The device includes some special features, namely:

- drive obtained avoiding any form of contamination of the inlet duct by fluids (oil and/or air) used to command the shutter. Thanks to this, when the pump is started again, the pumpdown to vacuum conditions is extremely fast as these contaminants are not present and no degassing is therefore required;
- suction flange maintained closed even when the pump is idle, so that pollutants cannot enter from the environment and no oil can overflow from the pump.

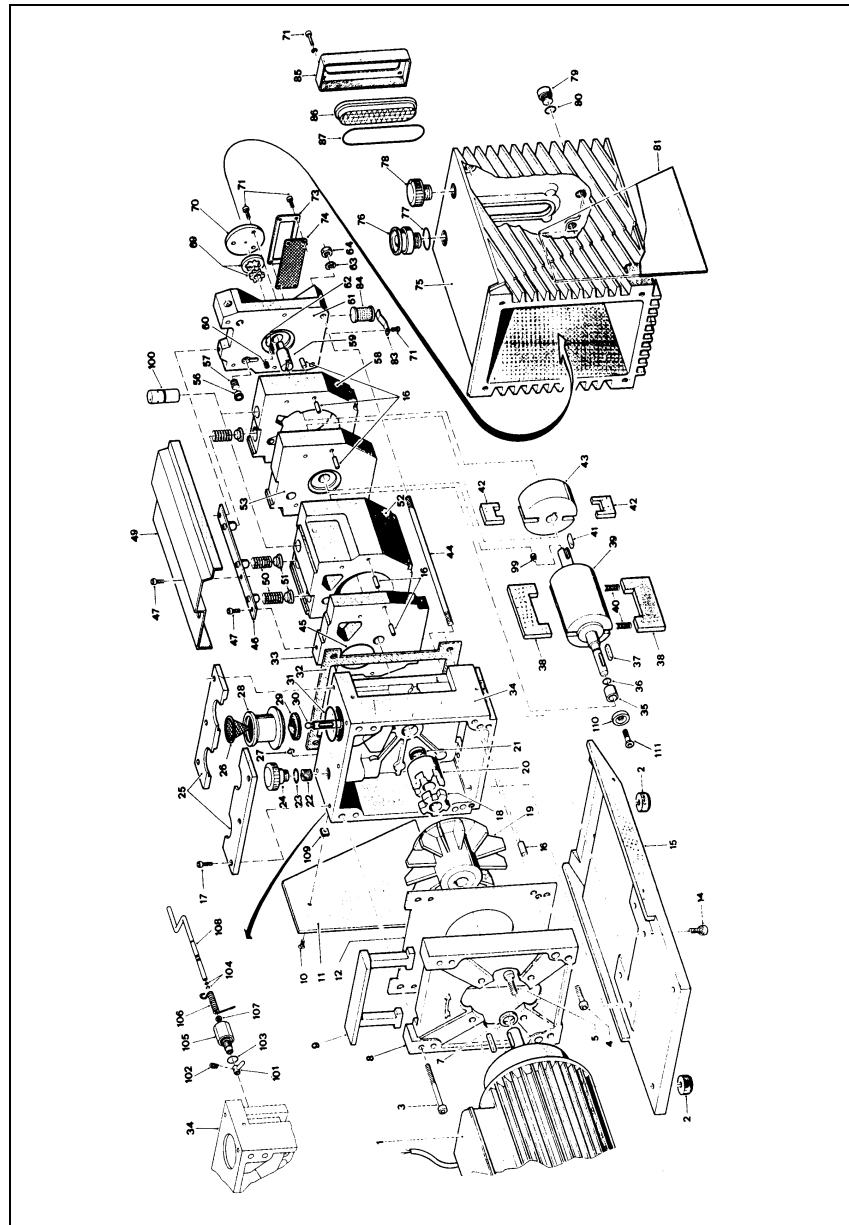


Figure 6 View of the pump



## 5 Technical Description

### Anti-Suckback Device

**Tab. 2**

| N° | DESCRIPTION                   | DS 102     | DS202      | DS302      | DS402      | DS602      |
|----|-------------------------------|------------|------------|------------|------------|------------|
| 1  | Electric motor                | SR03700702 | SR03700702 | SR03700702 | SR03700703 | SR03700703 |
| 1A | Electric motor<br>370-450W    | SR03700865 | SR03700865 | SR03700865 |            |            |
| 1B | Three phase<br>electric motor | SR03700922 | SR03700922 | SR03700922 | SR03700704 | SR03700704 |
| 2  | Rubber foot                   |            |            |            |            |            |
| 3  | Screw                         |            |            |            |            |            |
| 4  | Screw                         |            |            |            |            |            |
| 5  | Screw                         |            |            |            |            |            |
| 7  | Spacer                        |            |            |            |            |            |
| 8  | Motor support                 |            |            |            |            |            |
| 9  | Handle                        |            |            |            |            |            |
| 10 | Screw                         |            |            |            |            |            |
| 11 | Right sideplate               |            |            |            |            |            |
| 12 | Board                         |            |            |            |            |            |
| 14 | Screw                         |            |            |            |            |            |
| 15 | Support plate                 |            |            |            |            |            |
| 16 | Cylindrical pin               | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 17 | Screw                         |            |            |            |            |            |
| 18 | Rubber crown                  | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 19 | Half-joint with<br>fan        |            |            |            |            |            |
| 20 | Half-joint<br>(pump side)     |            |            |            |            |            |
| 21 | Oil-seal ring                 | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 22 | Special screw                 |            |            |            |            |            |
| 23 | OR gasket                     | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 24 | Gas Ballast<br>Valve knob     | SR03700219 | SR03700219 | SR03700219 | SR03700219 | SR03700219 |
| 25 | Half-plate                    |            |            |            |            |            |

| N° | DESCRIPTION           | DS 102           | DS202            | DS302            | DS402            | DS602            |
|----|-----------------------|------------------|------------------|------------------|------------------|------------------|
| 26 | Inlet screen          | SR03700237       | SR03700237       | SR03700237       | SR03700237       | SR03700237       |
| 27 | OR gasket             | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 28 | Suction flange        |                  |                  |                  |                  |                  |
| 29 | Anti-suckback shutter | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 30 | Piston                |                  |                  |                  |                  |                  |
| 31 | OR gasket             | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 32 | Casing gasket         | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 33 | Inner plate           | SR03700260       | SR03700260       | SR03700260       | SR03700362       | SR03700362       |
| 34 | Pump support          |                  |                  |                  |                  |                  |
| 35 | Anti-wear bushing     |                  |                  |                  | SR03700370       | SR03700370       |
| 36 | OR gasket             | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 37 | Kei motor side        | SR899998010<br>1 | SR899998010<br>1 | SR899998010<br>1 | SR899998030<br>1 | SR899998030<br>1 |
| 38 | First stage vane      | (1)              | (2)              | (2)              | (3)              | (3)              |
| 39 | First stage rotor     | SR03700273       | SR03700319       | SR03700319       | SR03700371       | SR03700371       |
| 40 | Spring                | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 41 | Tongue                | SR899998000<br>1 | SR899998000<br>1 | SR899998000<br>1 | SR899998040<br>1 | SR899998040<br>1 |
| 42 | Second stage vane     | (1)              | (2)              | (2)              | (3)              | (3)              |
| 43 | Second stage rotor    | SR03700269       | SR03700269       | SR03700269       | SR03700367       | SR03700367       |
| 44 | Tie-rod               | SR03700209       | SR03700308       | SR03700308       | SR03700335       | SR03700335       |
| 45 | OR gasket             | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |
| 46 | Bracket               |                  |                  |                  |                  |                  |
| 47 | Screw                 |                  |                  |                  |                  |                  |
| 49 | Valve cover plate     |                  |                  |                  |                  |                  |
| 50 | Valve spring          | (4) - (1)        | (4) - (2)        | (4) - (2)        | (5) - (3)        | (5) - (3)        |

## 5 Technical Description

### Anti-Suckback Device

| N°  | DESCRIPTION              | DS 102     | DS202      | DS302      | DS402      | DS602      |
|-----|--------------------------|------------|------------|------------|------------|------------|
| 51  | Valve gasket             | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 52  | First stage stator       | SR03700261 | SR03700316 | SR03700495 | SR03700363 | SR03700419 |
| 53  | Middle plate             | SR03700262 | SR03700262 | SR03700262 | SR03700421 | SR03700421 |
| 56  | GasBallast Valve shutter | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 57  | GasBallast Valve spring  | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 58  | Second stage stator      | SR03700321 | SR03700321 | SR03700321 | SR03700364 | SR03700364 |
| 59  | Oil pump joint           | SR03700280 | SR03700280 | SR03700280 | SR03700390 | SR03700390 |
| 60  | Special screw            |            |            |            |            |            |
| 61  | End plate                | SR03700285 | SR03700285 | SR03700285 | SR03700391 | SR03700391 |
| 62  | Tongue                   |            |            |            |            |            |
| 63  | Washer                   |            |            |            |            |            |
| 64  | Nut                      |            |            |            |            |            |
| 69A | External oil pump gear   | SR03700281 | SR03700281 | SR03700281 | SR03700281 | SR03700281 |
| 69B | Internal oil pump gear   | SR03700277 | SR03700277 | SR03700277 | SR03700277 | SR03700277 |
| 70  | Oil pump cover           |            |            |            |            |            |
| 71  | Screw                    |            |            |            |            |            |
| 73  | Plate                    |            |            |            |            |            |
| 74  | Filter                   |            |            |            |            |            |
| 75  | Casing                   |            |            |            |            |            |
| 76  | Exhaust flange           |            |            |            |            |            |
| 77  | OR gasket                | (4) - (1)  | (4) - (2)  | (4) - (2)  | (5) - (3)  | (5) - (3)  |
| 78  | Oil fill plug            | 03.701218  | 03.701218  | 03.701218  | 03.701218  | 03.701218  |
| 79  | Oil drain plug           | SR03700256 | SR03700256 | SR03700256 | SR03700256 | SR03700256 |
| 80  | Oil drain plug gasket    | SR03700256 | SR03700256 | SR03700256 | SR03700256 | SR03700256 |
| 81  | Left sideplate           |            |            |            |            |            |

| N°  | DESCRIPTION    | DS 102    | DS202     | DS302     | DS402     | DS602     |
|-----|----------------|-----------|-----------|-----------|-----------|-----------|
| 83  | Spring         |           |           |           |           |           |
| 84  | Filter         |           |           |           |           |           |
| 85  | Flange         |           |           |           |           |           |
| 86  | Sight glass    | 03.701326 | 03.701326 | 03.701326 | 03.701326 | 03.701326 |
| 87  | OR gasket      | (4) - (1) | (4) - (2) | (4) - (2) | (5) - (3) | (5) - (3) |
| 99  | Special screw  |           |           |           |           |           |
| 100 | Piston         |           |           |           |           |           |
| 101 | Arm            |           |           |           |           |           |
| 102 | Spring         | (1)       | (2)       | (2)       | (3)       | (3)       |
| 103 | OR gasket      | (4) - (1) | (4) - (2) | (4) - (2) | (5) - (3) | (5) - (3) |
| 104 | OR gasket      | (4) - (1) | (4) - (2) | (4) - (2) | (5) - (3) | (5) - (3) |
| 105 | Sleeve         |           |           |           |           |           |
| 106 | Spring         | (1)       | (2)       | (2)       | (3)       | (3)       |
| 107 | Ring           | (4) - (1) | (4) - (2) | (4) - (2) | (5) - (3) | (5) - (3) |
| 108 | Arm            |           |           |           |           |           |
| 109 | Insert         |           |           |           |           |           |
| 110 | Special washer |           |           |           |           |           |
| 111 | Screw          |           |           |           |           |           |

## NOTES

**Tab. 3**

|   |                                                            |
|---|------------------------------------------------------------|
| 1 | Part of DS 102 Major Maintenance Kit (P/N 9499380)         |
| 2 | Part of DS 202-302 Major Maintenance Kit (P/N9499381)      |
| 3 | Part of DS 402-602 Major Maintenance Kit (P/N 9499382)     |
| 4 | Part of DS 102-202-302 Minor Maintenance Kit (P/N 9499370) |
| 5 | Part of DS 402-602 Minor Maintenance Kit (P/N 9499371)     |

## Technical Data

The following table lists the main technical data of the DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph and DS602 3Ph pumps.

**Tab. 4**

| TECHNICAL DATA                            | Hz        | UNITS                     | DS102              | DS202              | DS302              | DS402              | DS602              |
|-------------------------------------------|-----------|---------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| FREE AIR DISPLACEMENT                     | 60        | l/min (cfm)               | 114 (4)            | 192 (6,8)          | 285 (10)           | 410 (14,5)         | 605 (21,4)         |
|                                           | 50        | l/min (m <sup>3</sup> /h) | 95 (5,7)           | 160 (9,6)          | 237 (14,2)         | 342 (20,5)         | 540 (30,2)         |
| PUMPING SPEED*                            | 60        | cfm                       | 3,5                | 5,8                | 8,2                | 12,3               | 18,3               |
|                                           | 50        | m <sup>3</sup> /h         | 5                  | 8,3                | 11,6               | 17,4               | 26                 |
| ULTIMATE PARTIAL PRESSURE*                |           | mbar                      | 10 <sup>-4</sup>   | 10 <sup>-4</sup>   | 10 <sup>-4</sup>   | 10 <sup>-4</sup>   | 10 <sup>-4</sup>   |
| ULTIMATE TOTAL PRESSURE*                  |           | mbar                      | 2·10 <sup>-3</sup> | 2·10 <sup>-3</sup> | 2·10 <sup>-3</sup> | 2·10 <sup>-3</sup> | 2·10 <sup>-3</sup> |
| ULTIMATE TOTAL PRESSURE WITH GAS BALLAST* |           | mbar                      | 2·10 <sup>-2</sup> | 2·10 <sup>-2</sup> | 2·10 <sup>-2</sup> | 1·10 <sup>-2</sup> | 1·10 <sup>-2</sup> |
| WATER VAPOR TOLERANCE                     |           | mbar                      | 15                 | 15                 | 20                 | 30                 | 30                 |
| WATER VAPOR CAPACITY                      |           | g/h                       | 60                 | 100                | 160                | 350                | 550                |
| OIL CAPACITY min/max                      |           | l                         | 0,3/0,5            | 0,4/0,6            | 0,4/0,6            | 0,6/1              | 0,6/1              |
| MOTOR RATING 1ph                          | 50/<br>60 | kW                        | 0.72               | 0.72               | 0.72               | 1.1                | 1.1                |

|                                      |    |     |                       |                       |                       |                       |
|--------------------------------------|----|-----|-----------------------|-----------------------|-----------------------|-----------------------|
| MOTOR RATING<br>Vac 3 ~              | kW |     |                       |                       |                       |                       |
| Y 380-415                            | 50 |     |                       | 0.6                   | 0.6                   |                       |
| Y 460                                | 60 |     |                       | 0.5                   | 0.5                   |                       |
| △ 200-230                            | 50 |     |                       | 0.5                   | 0.5                   |                       |
| △ 200-230                            | 60 |     |                       | 0.3                   | 0.3                   |                       |
| NOMINAL<br>ROTATION SPEED            | 50 | rpm | 1500                  | 1500                  | 1500                  | 1500                  |
|                                      | 60 | rpm | 1800                  | 1800                  | 1800                  | 1800                  |
| OIL<br>TEMPERATURE                   |    | °C  | 50                    | 50                    | 52                    | 70                    |
| (pump<br>operating)**                |    | °F  | 122                   | 122                   | 126                   | 158                   |
| Internal use Only                    |    |     |                       |                       |                       |                       |
| Max Altitude<br>2000m                |    |     |                       |                       |                       |                       |
| Installation<br>category             | II |     |                       |                       |                       |                       |
| Pollution degree                     | 2  |     |                       |                       |                       |                       |
| OPERATING<br>TEMPERATURE<br>RANGE    |    | °C  | 12 - 40               | 12 - 40               | 12 - 40               | 12 - 40               |
| WEIGHT                               |    | Kg  | 22                    | 24                    | 24                    | 36                    |
|                                      |    | lb  | 48                    | 53                    | 53                    | 79                    |
| INLET FLANGE                         | DN |     | 25KF                  | 25KF                  | 25KF                  | 25KF                  |
| EXHAUST FLANGE                       | DN |     | 25KF                  | 25KF                  | 25KF                  | 25KF                  |
| Dimensions (cf.<br>also next table): |    |     |                       |                       |                       |                       |
| - length, 1 ph                       | mm |     | 454                   | 491                   | 491                   | 530                   |
| - width                              | mm |     | 134                   | 134                   | 134                   | 164                   |
| - height                             | mm |     | 212                   | 212                   | 212                   | 242                   |
| Input single phase versions ***      |    |     |                       |                       |                       |                       |
| Nominal voltages:                    | 50 | V   | <u>100</u><br>200-230 | <u>100</u><br>200-230 | <u>100</u><br>200-230 | <u>100</u><br>200-230 |

# 5 Technical Description

## Technical Data

|                   | 60 | V | <u>100-120</u><br>200-230 | <u>100-120</u><br>200-230 | <u>100-120</u><br>200-230 | <u>100-120</u><br>200-230 | <u>100-120</u><br>200-230 |
|-------------------|----|---|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Maximum currents: | 50 | A | <u>8,8</u><br>5,4         | <u>8,8</u><br>5,4         | <u>8,8</u><br>5,4         | <u>11,8</u><br>6,2        | <u>11,8</u><br>6,2        |
|                   | 60 | A | <u>8,8</u><br>4,4         | <u>8,8</u><br>4,4         | <u>8,8</u><br>4,4         | <u>12,8</u><br>6,4        | <u>1,8</u><br>6,4         |

- \* According to PNEUROP 6602
- \*\* At ultimate total pressure, 20 °C (68 °F) room temperature
- \*\*\* For more details about the electrical motors see pages from 17 to 20

## Dimensions

Tab. 5

| MODEL        | UNITS | A    | B    | C    | D    | H   | L    | M    | N   | P   | R   | W   |
|--------------|-------|------|------|------|------|-----|------|------|-----|-----|-----|-----|
| <b>DS102</b> | mm    | 30   | 30   | 35   | 3    | 212 | 430  | 250  | 100 | 105 | 165 | 132 |
|              | inch  | 1,18 | 1,18 | 1,38 | 0,12 | 8,3 | 16,9 | 98   | 3,9 | 4,1 | 6,5 | 5,2 |
| <b>DS202</b> | mm    | 30   | 30   | 35   | 3    | 212 | 467  | 250  | 100 | 105 | 205 | 132 |
|              | inch  | 1,18 | 1,18 | 1,38 | 0,12 | 8,3 | 18,4 | 98   | 3,9 | 4,1 | 8,1 | 5,2 |
| <b>DS302</b> | mm    | 30   | 30   | 35   | 3    | 212 | 467  | 250  | 100 | 105 | 205 | 132 |
|              | inch  | 1,18 | 1,18 | 1,38 | 0,12 | 8,3 | 18,4 | 98   | 3,9 | 4,1 | 8,1 | 5,2 |
| <b>DS402</b> | mm    | 40   | 40   | 40   | 6    | 242 | 541  | 261  | 130 | 100 | 230 | 164 |
|              | inch  | 1,57 | 1,57 | 1,57 | 0,24 | 9,5 | 21,3 | 10,3 | 5,1 | 3,9 | 9,1 | 6,5 |
| <b>DS602</b> | mm    | 40   | 40   | 40   | 6    | 242 | 541  | 261  | 130 | 100 | 230 | 164 |
|              | inch  | 1,57 | 1,57 | 1,57 | 0,24 | 9,5 | 21,3 | 10,3 | 5,1 | 3,9 | 9,1 | 6,5 |

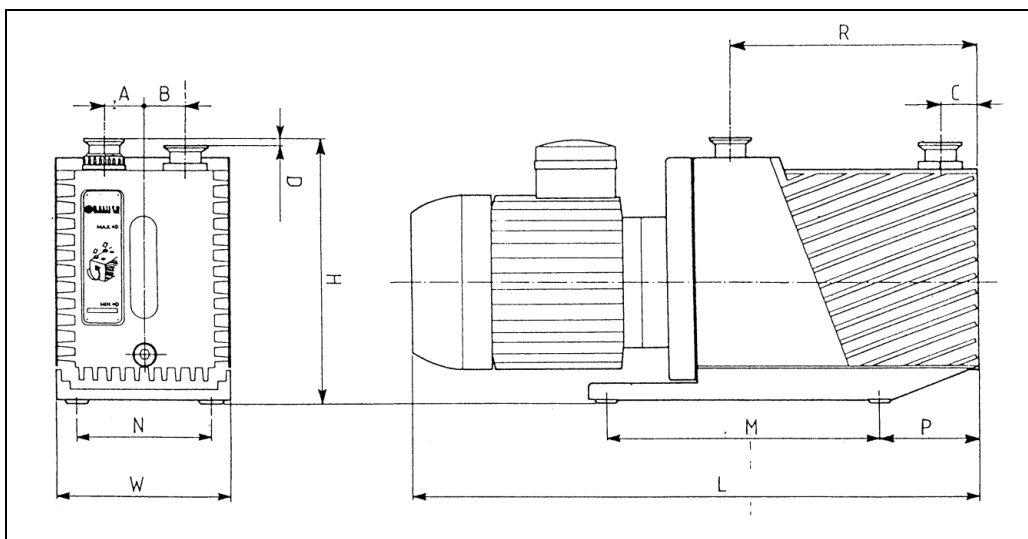
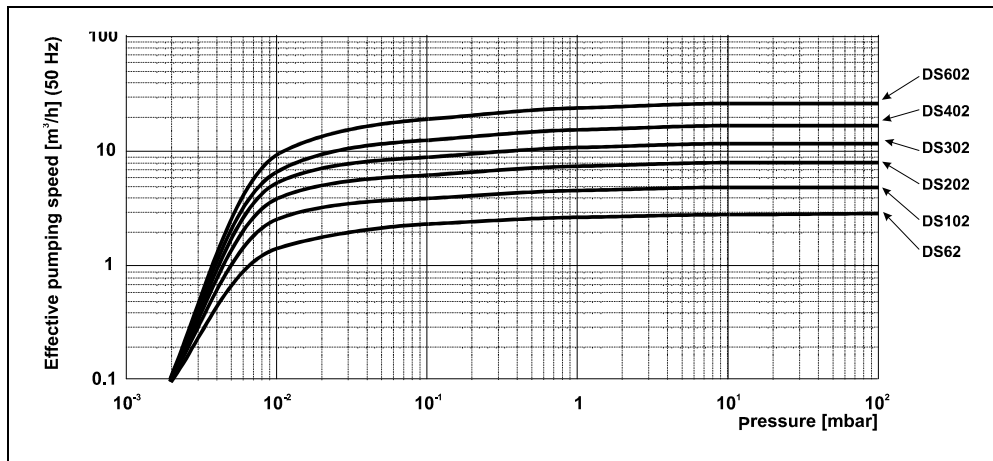


Figure 7 Pump dimensions



## 5 Technical Description

### Dimensions



**Figure 8** Typical Performance Curves

## Safety Precautions

- Always carry the pump by means of the handles provided.
  - The pump must be set in position taking the upmost care in order to avoid accidental falls.
- 

### **WARNING!**



**In case of a need to handle the pump after a period of operation, it must be left to cool first as the external surface temperature may be in excess of 60 °C.**

---

## Transport and Installation

The pumps are shipped to the customer inside cardboard boxes.

Total weight of the pack, including the pump, is between 30 Kg. and 40 Kg. according to the different pump models.

The case must be handled with care, using appropriate lifting equipment.

### **CAUTION!**

When moving the case, ensure that it is securely bound to the lifting equipment and that the equipment is strong enough to support the weight.

---

The pump's working environment is a traditional industrial environment. Naturally sites with corrosive vapors or excessive heat are best avoided.

Room temperature should ideally be between 12 °C and 40 °C.

If the temperature is not inside this range, consult Agilent technical service for the changes required.

Setting the pump in position should be performed as follows:

- Pump laid on the ground. There are no special instructions for this type of installation, except that the floor should be as flat as possible and suited to bear the weight of the pump (it should ideally be a concrete floor) and of any accessories mounted on it. Note that the pump is stable on its base plate and it should not be necessary to anchor it to the floor with bolts and screws; also vibrations to and from the pump are greatly reduced by the use of rubber feet.
- Pump off the ground. In this case, the user must design a suitable support structure, remembering the following points:
  - the plane supporting the pump must be perfectly horizontal;
  - the structure should be adequately rigid;
  - the relevant safety precautions should be applied.
- Note also that the pump should be attached to the supporting structure after replacing the rubber feet with special anti-vibration feet, which should be screwed to the pump base and to the supporting plane.

After taking the pump out of its packing case, you are advised to make the following checks:

Ensure that the pump has not suffered any damage during shipping.

Check that the guards are mounted correctly (side plates, refs. 11 and 81 in fig. *View of the pump*) and that there are no uncovered or loose parts.

## Preliminary Operations

Before starting the pump, fill up with lubricating oil as the pumps are delivered empty.

**NOTE**

A tin of oil is included in the packing.

---

For details on how to fill up, see Scheduled Maintenance Card 01.

**CAUTION!**

Oil must be poured into the casing through the special threaded plughole and NOT through the suction line.

---

**WARNING!**



Take out the protective caps on the suction and exhaust flanges before doing anything else. In the event of an accidental start-up, the air inside the pump could violently expel the protective caps and harm the operator.

---

# SECTION II

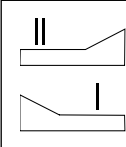
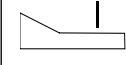
## Electrical motors

### Single Phase Universal Motors

They are world wide motors, dual voltages and dual frequencies, and they are in accordance with major international standards (UL, CSA, CE). On the table below are shown the technical specifications of the different motors.

**CAUTION!**

Before connecting to the mains, check the position of the voltage selector situated inside the electrical box (see next paragraph).

|                                                                                   |                     |                               |
|-----------------------------------------------------------------------------------|---------------------|-------------------------------|
|  | <b>HIGH VOLTAGE</b> | <b>200 to 230 V, 50/60 Hz</b> |
|  | <b>LOW VOLTAGE</b>  | <b>100 to 120 V, 50/60 Hz</b> |

**Figure 9** Voltage selector

The motor has a main on/off bipolar switch.

The motor includes a thermal protection against overloads. If the motor is switched off by this relay, it can be started again after a few minutes.

Consult the trouble shooting section in order to understand and remove the overload cause.

**NOTE**

Fuses installed as part of the building installation must be in accordance with the table below.

If you start a pump at low temperature, the current absorbed by the motors will be for several seconds higher than the nominal one, for this reason the fuse on the line must be of the slow-blow type.

**Tab. 6**

| PUMP                       | NOMINAL<br>VOLTAGE<br>[V] | HZ | RPM  | KW   | START UP<br>CURRENT [A] | FULL LOAD<br>CURRENT<br>[A] | MAXIMUM FUSE<br>RATING [A] |
|----------------------------|---------------------------|----|------|------|-------------------------|-----------------------------|----------------------------|
| DS 102<br>DS 202<br>DS 302 | 100                       | 50 | 1450 | 0.45 | 44                      | 8.8                         | 10                         |
|                            | 100 to 120                | 60 | 1730 | 0.55 | 42                      | 8.8                         | 10                         |
|                            | 200 to 230                | 50 | 1450 | 0.45 | 27                      | 5.4                         | 8                          |
|                            | 200 to 230                | 60 | 1730 | 0.55 | 24                      | 4.4                         | 6                          |
| DS 402<br>DS 602           | 100                       | 50 | 1450 | 0.75 | 75                      | 11.8                        | 16                         |
|                            | 100 to 120                | 60 | 1730 | 0.90 | 82                      | 12.8                        | 16                         |
|                            | 200 to 230                | 50 | 1450 | 0.75 | 43                      | 6.2                         | 8                          |
|                            | 200 to 230                | 60 | 1730 | 0.90 | 39                      | 6.4                         | 8                          |

## Single-Phase Universal Motors Voltage Setting

**CAUTION!**

Before connecting to the mains, check that your electrical mains voltage corresponds to the motor voltage setting.

---

**WARNING!**

**Disconnect the motor from the mains before opening the electrical box.**



---

The motors are factory set to operate at 200-230 V (50/60 Hz) nominal voltages.

To modify the voltage setting, change the position of the voltage selector situated inside the motor electrical box.

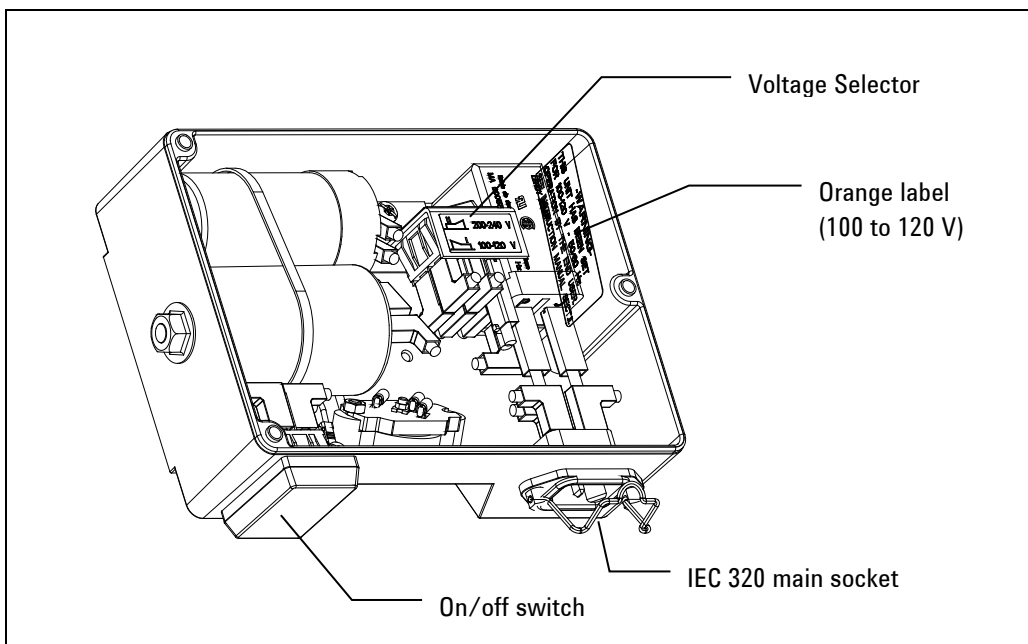
To confirm the change of the voltage setting from High Voltage to Low Voltage, glow the orange label above the yellow label positioned on the electrical box cover.

---

**WARNING!**

**Close the motor electrical box cover before connecting to the main.**





**Figure 10** Motor Electrical box

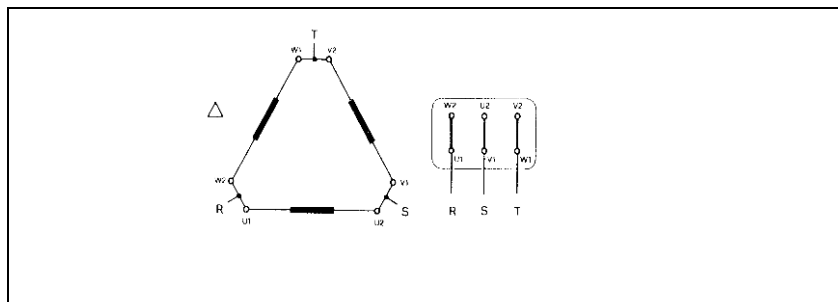


## Three Phase Universal Motors

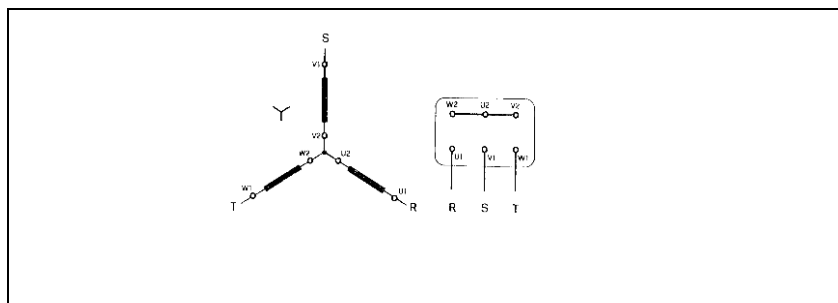
They are world wide motors in accordance with major international standards (UL, CSA, CE). On the table below are shown the technical specifications of the different motors.

### CAUTION!

Before starting the pump, give a short activation to the power switch, in order to check that the motor turns in the direction indicated by the arrow on the motor body. Should it rotate in the opposite direction, invert the connection of any two phases on the motor terminals.



**Figure 11** LOW VOLTAGE CONNECTION for power supply of 220V to 230 V 50 - 60 Hz DELTA CONNECTION



**Figure 12** HIGH VOLTAGE CONNECTION for power supply 380V to 460V 50 - 60 Hz STAR CONNECTION

## Electrical Connections of Three-Phase World Wide Motors

**Power supply cord:** The correct cable for electrical wiring is a four wires (R+S+T+Earth) cable. The wire section has to be at least 1.6 mm<sup>2</sup> (AWG14) 105°C AWG14.

To connect the motor to the three-phase mains, follow the wiring diagram on the motor plate or on the inside of the terminal strip.

Install a multipolar switch on the power line, with contact spacing of at least 3 mm. A magneto-thermal switch must be installed downstream of this switch, tripping at 130-140 % of the nominal current Type C.

**Tab. 7**

| CONNECTION | FREQ. [HZ] | NOMINAL VOLTAGES [V] | POWER [KW] | START UP CURRENT [A] | CURRENT [A] | MAX FUSE RATING [A] |
|------------|------------|----------------------|------------|----------------------|-------------|---------------------|
| STAR       | 50         | 380 to 415           | 0.75       | 15                   | 2.6         | 4                   |
| STAR       | 60         | 460                  | 0.90       | 14                   | 2.4         | 4                   |
| DELTA      | 50         | 200 to 230           | 0.75       | 26                   | 4.4         | 6                   |
| DELTA      | 60         | 200 to 230           | 0.90       | 21                   | 4.8         | 6                   |

### CAUTION!

It is recommended to connect the pump to the power supply through a dedicated switch on the main electrical panel of the installation, or in proximity of the power supply connection point the circuit breaker has to be type C.

### CAUTION!

In order to easily identify which disconnecting device related is used for the DS Pump, the Circuit Breaker must be marked to Marked.

## 5 Technical Description

### Connections to the Inlet and Exhaust Flanges

#### CAUTION!

Only electrically trained staff may install the 3Ph models and may change the connection from STAR to DELTA or from DELTA to STAR.

---

Before starting the pump, give a short activation to the power switch, in order to check that the motor turns in the direction indicated by the arrow on the motor body. Should it rotate in the opposite direction, invert the connection of any two phases on the motor terminals.

One of the most effective method to verify the rotational way of the motor, it is to start the pump with a sheet of paper on the pump inlet. If the paper is sucked by the pump then the motor runs in the right way.

## Connections to the Inlet and Exhaust Flanges

Remove the protective caps from both flanges. Remove protective cap from inlet flange. Remove metallic cap and rubber membrane from exhaust flange as explained in the Installation chapter. Connect the system to be evacuated to the inlet flange, using a centering ring with OR and a locking collar.

#### NOTE

For guaranteed reliable sealing, use an OR gasket in Perbunan or Viton.

---

The inlet duct is equipped with a sieve filter preventing solid particles from entering and damaging the pump.

#### NOTE

When the gases to be pumped out contain dust, it is advisable to insert a dust filter before the inlet flange.

---

**NOTE**

When the gases to be pumped out contain large quantities of vapor, it is advisable to include a condense separator before the inlet flange.

---

To make best use of the pump's capacity, use only short, straight piping, with a diameter not smaller than that of the inlet flange.

**NOTE**

If rigid piping is used, it is good practice to use a flexible joint in order to avoid undue forcing of the connection on the pump.

---

The exhaust duct must be connected to a pipe that will take away the pumped out gases.

**NOTE**

Application of an oil trap filter is necessary to avoid pollution of the surrounding atmosphere by the oil present in the exhaust duct during pump operation.

---

**CAUTION!**

Never block the pump exhaust line. This would cause overpressure in the casing with the risk of breaking the glass window of the level indicator and/or expelling the oil seal gasket.

---

## Starting and Running the Pump

Before being put into service, in order to reach maximum vacuum, the pump must be left running for about an hour with the gas ballast valve open. This will eliminate any humidity from the oil.

## 5 Technical Description

### Starting and Running the Pump

---

#### **WARNING!**



The pump is designed for operation with neutral or non-corrosive fluids. It is absolutely forbidden to use potentially explosive or flammable substances.

---

#### **NOTE**

If the pump is started with cold oil, initially more than normal noise will be heard; this will last for a few minutes only until the oil reaches its working temperature.

---

There are no special instructions for normal operation of the pump, which is delivered to you after completion of a running-in cycle in the factory. The maintenance operations for a pump in working condition may be found on the Scheduled Maintenance Cards.

If the pump is sucking in environments with liquids or vapors, it is important to leave it to run with the gas ballast open so as to avoid condensation of the vapors and contamination of the lubricating oil as a result. When there is no danger of the oil being contaminated, the gas ballast valve should be closed to reduce the pump's power absorption.

The gas ballast valve will prove useful in the cold season when, due to the lower room temperatures, it takes longer for the pump to reach rated temperature when switched on. Compression of the air forced in by the gas ballast valve develops heat, which in turn increases temperature of the oil faster.

#### **NOTE**

If gases without vapors are sucked in, you are advised to open the gas ballast device from time to time in order to eliminate the traces of humidity they contain.

---

**NOTE**

For repetitive work cycles, with brief time intervals in between, it is better not to stop the pump.

---

## Stopping the Pump

There are no special procedures for switching the pump off; it needs only to be disconnected from the electric power by means of the bipolar switch (version with single-phase motor - see section **MODELS WITH SINGLE-PHASE UNIVERSAL MOTORS**) or of the multipolar switch (version with three-phase motor - see section **MODELS WITH THREE-PHASE UNIVERSAL MOTORS**).

When the pump is stopped, the anti-suckback device makes it possible to maintain vacuum in the vessel connected on the inlet flange of the pump.

If the pump is expected to be stopped for a lengthy period, or in any case if it has pumped in large amounts of vapors, it is good practice to run it with the gas ballast open and the inlet line closed for a few minutes before switching off in order to limit the risk of corrosion or scaling due to pollution of the oil by condensed vapors.

## Safety Rules

Personnel responsible for pump operation and maintenance must be well-trained and must be aware of the accident prevention rules.

The accident prevention precautions contained in this section must be respected at all times during operation and maintenance of the pump to avoid damage to operators and to the pump.

These precautions are provided in the form of **WARNING** and **CAUTION** notes.

---

**WARNING!**

Operating procedures, technical information and precautions which, if not respected and/or implemented correctly may cause body harm to operators.

---

**CAUTION!**

Operating procedures, technical information and precautions which, if not respected and/or implemented correctly may cause damage to the pump.

---

## Warning Notes

- a Death may result from contact with high voltages. Always take extreme care and observe the accident prevention regulations in force.
- b Always disconnect the power supply to the pump before maintenance work. Place a special warning signs over the power supply breaker switch: MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON.
- c If you are performing maintenance after the pump has been operating for a considerable time, allow sufficient time for it to cool as the external surface temperature may be in excess of 60 °C.
- d Failure to provide the pump with an earth connection may cause serious damage to operators. Always ensure that there is an earth connection and that it complies with the standards.
- e When cleaning the pump and its component parts, avoid the use of flammable or toxic solvents, such as benzin, benzol, ether or alcohol. The recommendation is to use a soap and water solution, preferably in ultrasound washing machines, taking care to dry all the cleaned parts at temperatures under 100 °C in order to eliminate residual moisture.
- f Prolonged overloads or breakdowns may cause the electric motor to overheat, and to release noxious smoke; remove the power immediately as a precaution and do not approach the pump at least until you have provided ventilation to drive out the smoke. Take care not to breathe in the fumes remaining inside the pump in the course of repair work.

- g** In case of fire, do not throw water on the pump. Switch the power off and use CO2 extinguishers.
- h** Carefully inspect the flanges to ensure that there are no dust, oil, dirt or defects of the mating surfaces, before making the required connections.
- i** Ensure that all joints and couplings are locked correctly before starting the pump again after repair work.
- j** Do not wear any objects that may become entangled in the mechanisms and/or act as conductors (chains, bracelets, etc.).
- k** Ensure that the tools to be used are in perfect working condition and have insulating grips, where necessary. Check that the insulating material of the cables and that the conductors of the test equipment do not show any signs of damage.
- l** Do not replace the oil immediately after stopping the machine as the oil may still be at high temperature.
- m** Perform repairs in clean and, where possible, dust-free areas. Protect all the clearances of connection points with suitable plastic caps and cover the machined surface areas of all parts stripped down until they are put back on the pump again.
- n** Do not start the pump if the protective sideplates (refs. 11 and 81 of fig. View of the pump) are not properly in place.



## Caution notes

- a** Before putting the pump back into operation after a breakdown, inspect it and check carefully for any other signs of damage.
- b** Before putting the pump back into operation after a breakdown, inspect it and check carefully for any other signs of damage.
- c** Use only tools that are in perfect working order and specially designed for the job; use of inappropriate or ineffective tools may cause serious damage.
- d** Perform repairs in clean and, where possible, dust-free areas. Protect all the clearances of connection points with suitable plastic caps and cover the machined surface areas of all parts stripped down until they are put back on the pump again.
- e** Always check the lubricant and that it is properly distributed through the pump; inadequate lubrication may damage the pump seriously.
- f** Give the parts some form of marking as you strip them down to ensure that you reassemble them again in the proper order.
- g** Check that there are no scratches or grooves on the machined shafts, in their seats inside the pump or on machine-ground surfaces. Slight scratches and abrasions may be eliminated with very fine emery paper or by a little light grinding.
- h** Before putting a group together, always spread a little oil over inner parts and mating surfaces. Replace all seals with original spare parts before reassembling components.

## Maintenance Actions

Maintenance may be seen as the totality of all scheduled and unscheduled maintenance work.

Maintenance includes:

- 1 **SCHEDULED MAINTENANCE: Maintaining the nominal state of operation.**
- 2 **UNSCHEDULED MAINTENANCE: Restoring the nominal state of operation**

### NOTE

The frequency with which repairs are performed depends on the process and presence of substances that shorten pump life (dust, abrasives, solvents, water, chemically aggressive substances).

---

The pump must be cleaned at regular intervals of time.

### CAUTION!

Do not clean with Alcohol the plastic or rubber components of the pump.

---

Use only the strictly necessary amount of lubricant; an excess of lubricating oil, like when there is none, may sometimes compromise proper operation of the pump.

Only the recommended lubricants, or lubricating oils with similar characteristics and known and experimented quality, should be used.

Oil changes must be made with the oil at a sufficiently high temperature, after leaving the pump to cool for a few minutes following operation.

The drain and filler plugs must not be left open any longer than is strictly necessary.

When performing maintenance, look out for all signals that may precede a breakdown, in particular:

- traces of corrosion;
- oil leaks;

## 5 Technical Description

### Maintenance Actions

- slack joints or couplings.

Maintenance technicians must:

- be aware of all applicable national directives concerning accident prevention during work on motor-driven pumps and should know how to apply them;
- have read and understood all the sections on "Safety Rules";
- be familiar with the essential design features and operation of the pump;
- know how to use and consult the pump documentation;
- be concerned about proper operation of the pump;
- make a note of any irregularities in operation of the pump and take the necessary action, where appropriate.

Use original spare parts wherever possible and repair a broken part as best as possible on site or send it back to the manufacturer for repairs.

For all problems arising, or to order spare parts, refer to our service department.

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## Lubricants

It will be readily understood how important adequate lubrication is to high technology pumps like the Agilent vacuum pumps. Correct use of appropriate lubricants makes a significant contribution to achieving best performance and warding off defects.

When handling lubricants, the following sanitary protection measures should be observed at all times:

- Avoid prolonged, excessive or repeated contact of the skin with products for lubrication, and also avoid directly inhaling the fumes or vapors of such products.
- Protect the skin by wearing appropriate clothes and equipment (e.g. special suits, glasses or, where permitted by the safety regulations, gloves) or by applying a special protective product.
- Clean the skin carefully after contact with the lubricants by washing freely with water and soap.
- Apply a skin cream after washing.
- Take off and change clothes or shoes on which oil has been spilled.
- Never put rags dripping with oil into the pockets of your clothes.

When disposing of waste lubricants, observe the following environment protection regulations:

- The lubricants risk contaminating the water and the ground! Therefore never pour lubricating products on to the ground, into water or in the sewage system. All violations of these rules are liable to persecution as provided for by law. When using lubricants always keep an oil can nearby.
- Take care in draining off waste oils. In disposal of these products respect all regulations in force concerning waste oil disposal.

The recommended lubricating oil is the Agilent Rotary Vane Fluid DS19 Type. The Rotary Vane Fluid DS19 Type is a general purpose mechanical pump fluid specifically engineered to provide superior performance in high speed direct drive mechanical pumps.

## 5 Technical Description

### Lubricants

These precisely distilled fluids (100 % solvent refined neutral paraffinic oil) deliver lower base pressure capability, faster pump-down cycles, and reduced maintenance requirements on both the pump and the fluid.

It is absolutely necessary to continue using the lubricants initially used to fill the tank. If this is not possible for organizational or business reasons, use only products with the same characteristics as the previous oils.

Only use of lubricants of suitable quality will guarantee safe operation of the pumps.

#### CAUTION!

Mineral oils and the PFPE oil are incompatible. To change from one type to another, the pump must be stripped down completely and all parts washed carefully to eliminate all oil residues.

---

If you expect to have to use other lubricants, first find out if the two products are compatible. In cases of doubt, the lubricant used up to that time must be flushed out by way of a pump flushing procedure.

#### CAUTION!

To avoid the risk of contaminating the oil, absolute cleanliness of the pump and surrounding area must be ensured during the lubrication procedures.

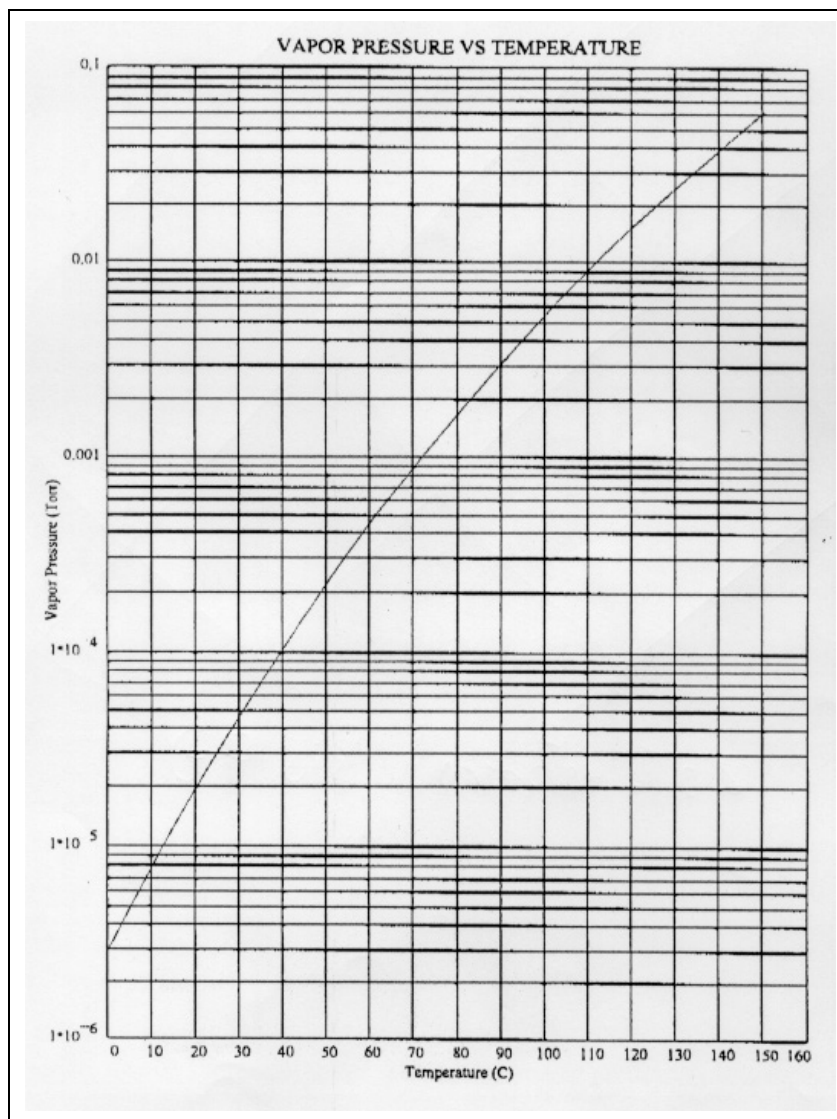
---

**Tab. 8**

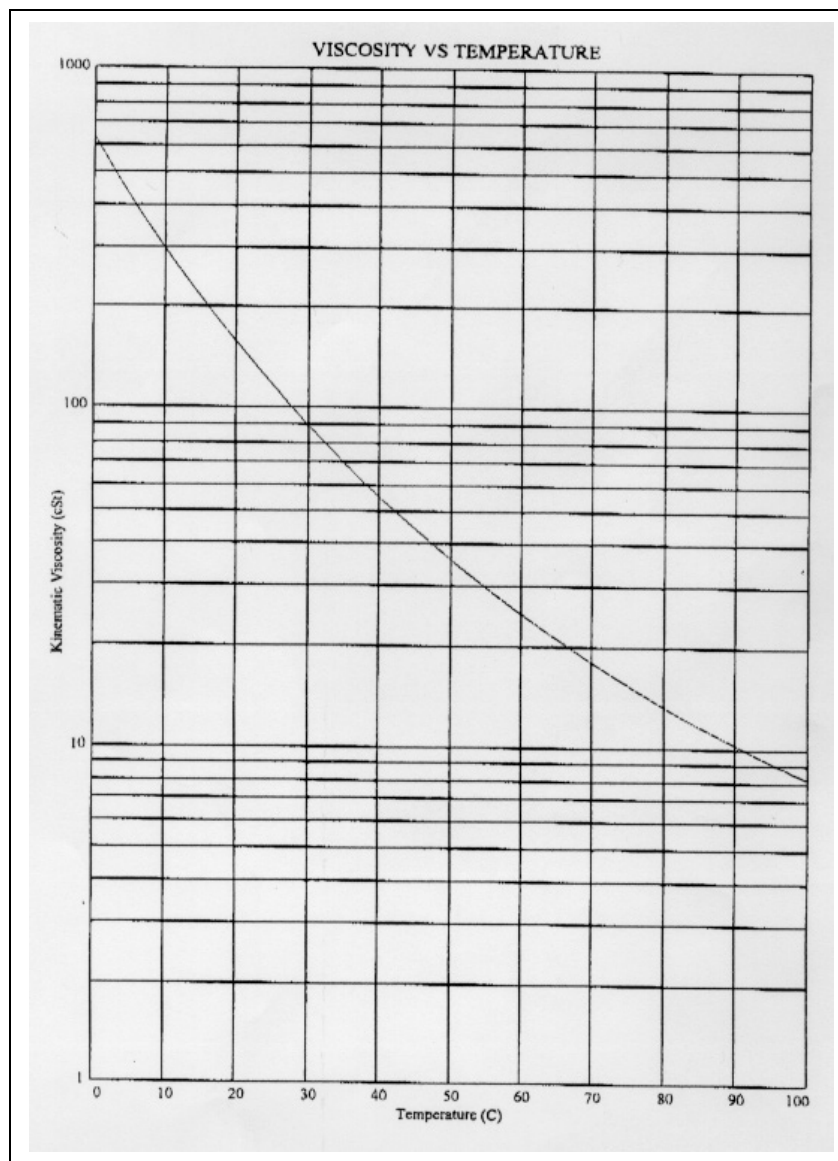
| PROPERTY                  | UNIT OF MEASURE | ROTARY VANE FLUID DS19<br>Type |
|---------------------------|-----------------|--------------------------------|
| Vapor pressure @ 25 °C    | Torr            | 1·10 <sup>-5</sup>             |
| Boiling point @ 0,01 torr | °C (°F)         | 112 (233)                      |
| Viscosity @ 40 °C         | cSt (SUS)       | 55 (258)                       |
| Viscosity @ 100 °C        | cSt (SUS)       | 8,1 (52,7)                     |
| Pour point                | °C (°F)         | -15 (5)                        |
| Flash point               | °C (°F)         | 213 (415)                      |
| Fire point                | °C (°F)         | 244 (472)                      |
| Density @ 25 °C           | g/ml            | 0,87                           |

## 5 Technical Description

### Lubricants



**Figure 13** Agilent Rotary Vane Fluid DS19 Type vapor pressure curve



**Figure 14** Agilent Rotary Vane Fluid DS19 Type vapor pressure curve



## Scheduled Maintenance Cards

**Tab. 9**

| No. | FREQUENCY       | PUMP STATUS (*) | OPERATION                 |
|-----|-----------------|-----------------|---------------------------|
| 01  | MONTHLY         | R               | OIL LEVEL CHECKING        |
| 02  | TWICE<br>YEARLY | S               | CLEANING THE INLET FILTER |

(\*) R=RUNNING   S=STOPPED

**NOTE**

When the pump is equipped with:

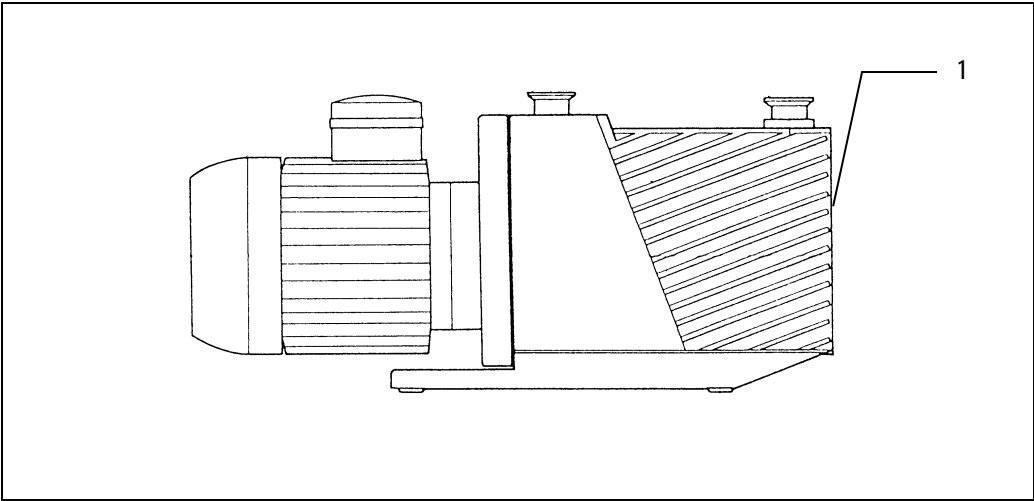
- a filter for dust on the inlet side
- a condensation separator on the inlet side

these items must also be maintained, by following the Manufacturer's instructions.

**NOTE**

The numbers used in the drawings on the Scheduled Maintenance Cards are those used in Figure View of the pump.

| Scheduled Maintenance |                                                                             | Card No.                                                                                              |
|-----------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|                       |                                                                             | 01                                                                                                    |
| Pump type:            | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                                                                                                       |
| Frequency: Monthly    | Time: 5 min                                                                 | Machine status: S  |
| Type of action:       | LUBRICATION                                                                 |                                                                                                       |
| Item:                 | LUBRICATING OIL 9499390                                                     |                                                                                                       |



| No. | Description                                  | Time  | Equipment                                                                              | Materials       |
|-----|----------------------------------------------|-------|----------------------------------------------------------------------------------------|-----------------|
| 1   | Check the oil level and top up if necessary. | 5 min | <ul style="list-style-type: none"><li>Cleaning rags</li><li>Liquid detergent</li></ul> | Oil P/N 9499390 |

**WARNING!**

**Machine powered: take care on account of moving parts and high voltages.**



**WARNING!**

**If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.**



1. Check the oil level through the sight glass: there are two clearly visible arrows indicating maximum and minimum levels for proper pump operation.

**NOTE**

The amount of oil consumed depends on the type of work, particularly on the volume being evacuated and on the frequency of the vacuum cycles.

---

2. If necessary, open the pump filling plug on the top plane of the pump and add oil.

**NOTE**

Before starting the procedure, check that the quality and type of oil used corresponds to the indications of Table Agilent Rotary Vane Fluid DS19 Type characteristics of this manual.

---

**CAUTION!**

Take great care to avoid oily residues or dust from entering the pump and jeopardizing proper operation.

---

3. Close the fill plug and, using cotton or other suitable material rags, clean thoroughly the outer surface of the top plane of the pump.

**CAUTION!**

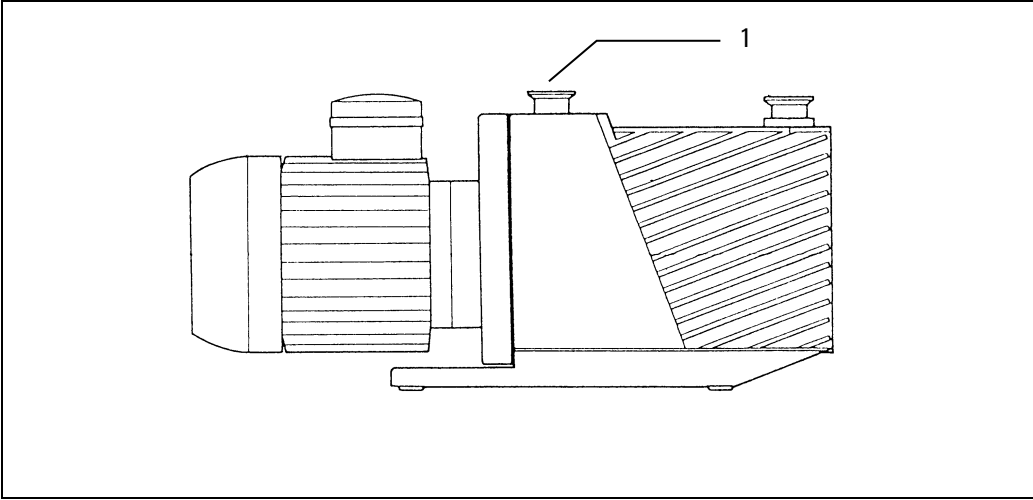
Never mix lubricants of different brands or quality. To change from one oil to another, the pump must be stripped down and all parts washed carefully, taking care to eliminate oil residues.

---

4. Open the gas ballast valve to purify the oil.

**5    Technical Description**  
Scheduled **Maintenance Cards**

| Scheduled Maintenance          |                                                                             | Card No.                          |
|--------------------------------|-----------------------------------------------------------------------------|-----------------------------------|
|                                |                                                                             | 02                                |
| <b>Pump type:</b>              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                                   |
| <b>Frequency:</b> Twice Yearly | <b>Time:</b> 10 min                                                         | <b>Machine status:</b> <b>S</b> R |
| <b>Type of action:</b>         | CLEANING                                                                    |                                   |
| <b>Item:</b>                   | INLET FILTER                                                                |                                   |



| No. | Description                                  | Time   | Equipment                                                                                  | Materials       |
|-----|----------------------------------------------|--------|--------------------------------------------------------------------------------------------|-----------------|
| 1   | Check the oil level and top up if necessary. | 10 min | <ul style="list-style-type: none"><li>▪ Cleaning rags</li><li>▪ Liquid detergent</li></ul> | Oil P/N 9499390 |

**WARNING!**



**Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.**

---

**WARNING!**



**If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.**

---

1. Remove the filter (26) from the inlet flange (28).
2. Clean the filter using an aqueous solution and dry with hot air.
3. Put the filter back in place.

**CAUTION!**

The filter element is designed to prevent solid particles in the air from entering the inside of the pump and if it is not working properly, serious damage could be caused the pump's inner parts. In such cases, it may be worthwhile installing a large surface area filter in the pump's suction line to improve effectiveness and reduce maintenance frequency.

If the pump sucks in gases and vapors that contain large quantities of solid particles, the filter must be checked and cleaned more often than indicated here.

---

## 5 Technical Description

### Scheduled Maintenance Cards

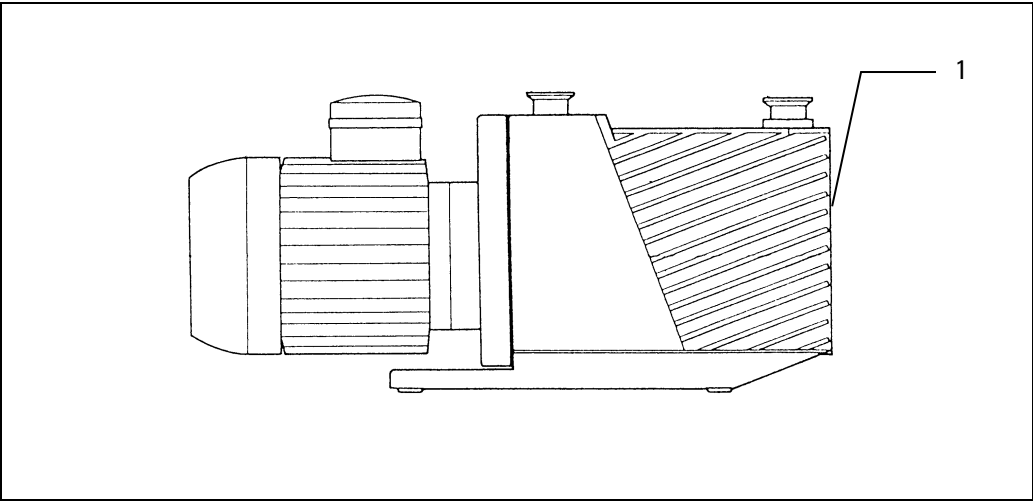
**Tab. 10** UNSCHEDULED MAINTENANCE CARDS

| No. | SUBJECT                                     | FREQUENCY          |
|-----|---------------------------------------------|--------------------|
| 101 | General Warnings                            |                    |
| 102 | Changing the oil                            | Whenever necessary |
| 103 | Disassembly of oil level indicator          | When damaged       |
| 104 | Disassembly of anti-suckback device shutter | When damaged       |
| 105 | Disassembly of electric motor and joint     | When damaged       |
| 106 | Disassembly of discharge valve              | When damaged       |
| 107 | Disassembly of oil circulating pump         | When damaged       |
| 108 | Disassembly of anti-suckback valve          | When damaged       |
| 109 | Disassembly of pump body                    | When damaged       |
| 110 | Replacement of oil seal ring/wear bushing   | When damaged       |
| 111 | Disassembly of functional block rotors      | When damaged       |

#### NOTE

The numbers used in the drawings on the Unscheduled Maintenance Cards are those used in Figure View of the pump.

| Unscheduled Maintenance        |                                                                             | Card No.<br>101        |
|--------------------------------|-----------------------------------------------------------------------------|------------------------|
| <b>Pump type:</b>              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                        |
| <b>Frequency:</b> Twice Yearly | <b>Time:</b> 10 min                                                         | <b>Machine status:</b> |
| <b>Type of action:</b>         | GENERAL WARNINGS                                                            |                        |
| <b>Item:</b>                   |                                                                             |                        |



1. Pump disassembly and reassembly require some basic operations and use of tools found in most workshops.



## 5 Technical Description

### Scheduled Maintenance Cards

#### NOTE

The pump's component parts are modular and therefore fully interchangeable. Whenever parts have to be replaced, please specify in the order:

Pump type

Serial number (marked on pump identification plate).

Reference number on the detailed diagram, name and part number as indicated on the spare parts and accessories list (section SPARE PARTS LIST of this manual).

To simplify maintenance work, two kits have been devised.

The Gaskets Kit, containing all the gaskets recommended for substitution after disassembly of the pump and the most commonly used removable parts.

You are advised to keep this kit in a dry room, away from heat and light (sun and UV rays) in order to prevent hardening of the elastomer parts.

The Maintenance Kit, including, as well as the Gaskets Kit, the 1st and 2nd stage vanes and other parts that it may be necessary to replace when overhauling the pump completely. These parts have a life of approx. two years under conditions of normal operation.

---

Tab. 11

| KIT             | Pump Type |           |           |                        |                        | ORDER CODE |
|-----------------|-----------|-----------|-----------|------------------------|------------------------|------------|
|                 | DS102 1Ph | DS202 1Ph | DS302 1Ph | DS402 1Ph<br>DS402 3Ph | DS602 1Ph<br>DS602 3Ph |            |
| GASKETS KIT     | X         | X         | X         |                        |                        | 9499370    |
| GASKETS KIT     |           |           |           | X                      | X                      | 9499371    |
| MAINTENANCE KIT | X         |           |           |                        |                        | 9499380    |
| MAINTENANCE KIT |           | X         | X         |                        |                        | 9499381    |
| MAINTENANCE KIT |           |           |           | X                      | X                      | 9499382    |
| GASKETS KIT     | X         | X         | X         |                        |                        | 9499370    |

- Before reassembling the pump, all parts must be cleaned thoroughly with a soap and water solution and rinsed with demineralized water.

**CAUTION!**

Particular care must be taken in cleaning the lubrication ducts and oil passage nozzles, which must always be kept free of impurities.

**CAUTION!**

Take care that the soap and water solution does not remain in the various cavities and recesses of the pump, in the blind holes in particular.

- Then dry the parts, by heating where possible but without using compressed air (it generally contains polluting substances), taking care not to exceed a temperature of 80 °C.
- The sequence of reassembly is the inverse of the disassembly operation. Take the same precautions:

## 5 Technical Description

### Scheduled Maintenance Cards

- Spread some clean oil of the same type as used to lubricate the pump over the different parts, in particular the inner lip of the oil seal, the bushings and vanes.
- If the oil seal has the dust-block lip, 2/3 fill the area between the two lips with ISOFLEX TOPAS NB5051 grease.

#### CAUTION!

Do not allow the grease into the area of the circumferential spring.

---

- Lock without forcing the nuts on the four tie-rods (44) connecting the parts of the pump block. Before locking, check that all parts are touching as required.
- Before putting the springs and shutters of the discharge valves back in, pour a little oil into the stators through the holes in the valves.

#### CAUTION!

Before hooking the recall spring to the anti-suckback command bar, make sure you have put the bar all the way into the sleeve of the rotary feedthrough to avoid bending the tapered part of the bar.

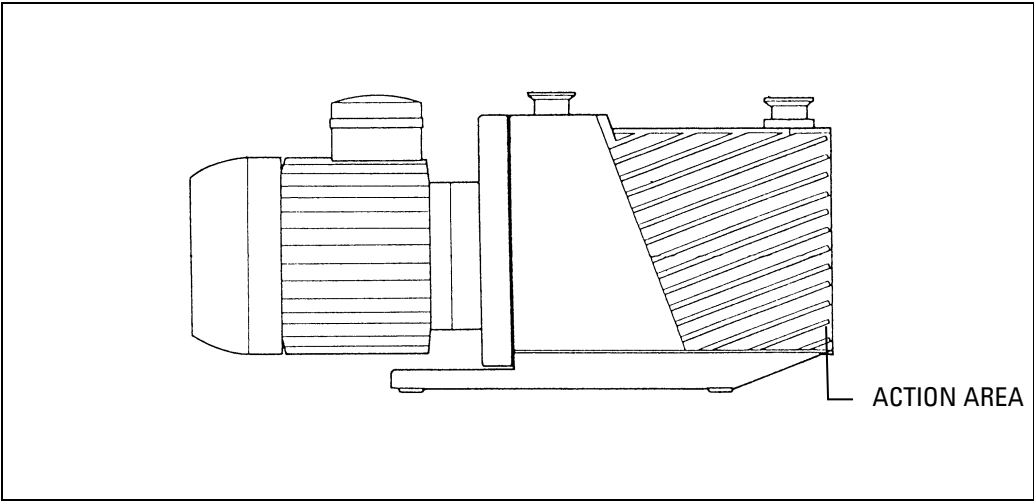
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#### NOTE

On DS102 1Ph, DS202 1Ph and DS302 1Ph pumps, fit the joint on the motor shaft after careful removal of grease from the surfaces and application of LOCTITE 242.

---

| Unscheduled Maintenance |                                                                             | Card No.<br>102            |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: AS REQUIRED  | Time: 30 min                                                                | Machine status: <b>S</b> R |
| Type of action:         | CHANGING OIL                                                                |                            |
| Item:                   | LUBRICANT 9499390                                                           |                            |



| Materials required:                                                                                       | Equipment required:                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Agilent Rotary Vane Fluid DS19 Type (1 liter) P/N 9499390</li></ul> | <ul style="list-style-type: none"><li>Fixed wrench for oil exhaust.</li><li>Tank to collect oil.</li><li>Cotton (or other suitable material) rags for cleaning.</li><li>Soap and water solution for cleaning.</li></ul> |

## 5 Technical Description

### Scheduled Maintenance Cards

---

#### **WARNING!**



Do not change the oil immediately after stopping the machine as the oil temperature may still be high.

---

---

#### **WARNING!**



Machine safety: disconnect all power lines. Hang the card "MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON" from the main breaker switch. When finished, remove the safety warning.

---

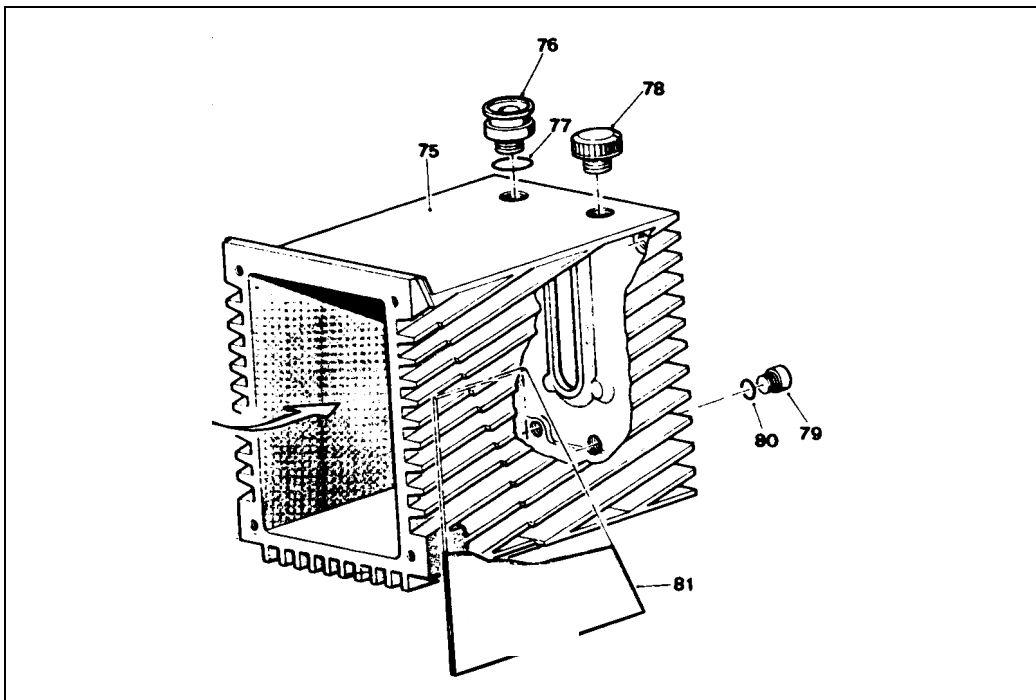
---

#### **WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.

---



**NOTE**

The oil has to be replaced:

1. when it has become particularly dirty due to the presence of contaminants in the environments being evacuated;
2. when the degree of vacuum achieved by the pump is no longer satisfactory, even after running for one hour with the gas ballast device open.

1. Place a tank for the waste oil under the oil drain plug (79).

## 5 Technical Description

### Scheduled Maintenance Cards

#### NOTE

You are advised to empty the pump before it cools down completely so that the oil flows out more easily.

---

#### NOTE

The pump contains up to 1 liter of oil - make sure the tank you use for the waste oil can hold this amount.

---

2. Open the oil drain plug (79) slowly; let the oil flow out by opening the fill plug (78) located at the top of the pump. In the meantime, clean the inside of the drain plug (magnetic plug).
3. After closing the drainage hole with its plug, make the pump perform a number of turns, giving it a brief current pulse, so as to let residual oil in the pump chambers flow out. Then drain off the residual oil again.
4. After all the oil has been drained, close the drain plug tight; take the tank of waste oil away and clean thoroughly, using rags of cotton or other suitable material to dry.
5. Clean thoroughly the pourer lip of the new oil tin; pour the oil slowly into the pump through the filler plug (78).

#### NOTE

Monitor through the sight glass at the front of the pump that you are not exceeding the maximum oil level permitted.

---

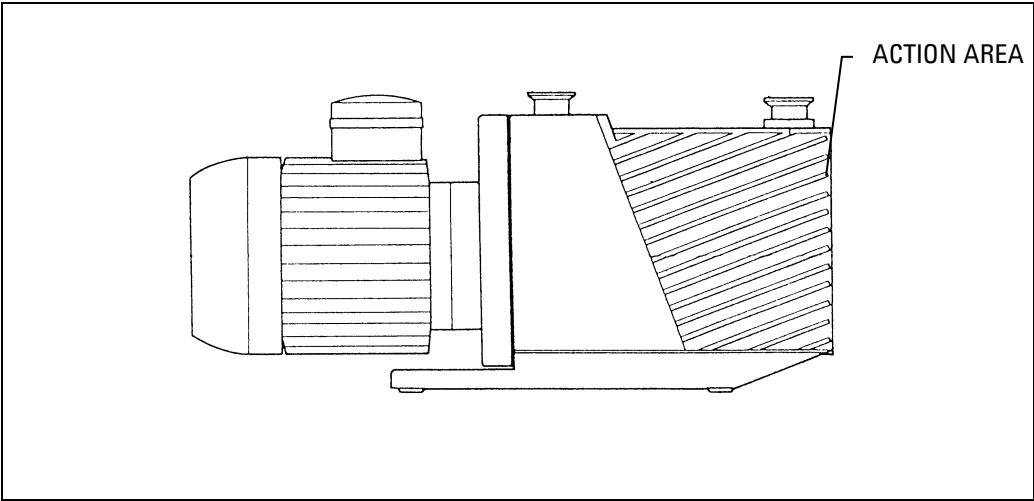
#### CAUTION!

For disposal of waste oil, it is strictly necessary to respect the currently applicable legislation on this matter.

---

6. To achieve a good vacuum level again, run the pump for one hour with the gas ballast open.

| Unscheduled Maintenance |                                                                             | Card No.<br>103            |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 30 min                                                                | Machine status: <b>S</b> R |
| Type of action:         | DISASSEMBLY                                                                 |                            |
| Item:                   | SIGHT GLASS                                                                 |                            |



| Materials required:                                                                                                                                                                                                                             | Equipment required:                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>OR gaskets<br/>(for pumps DS102 1Ph, DS202 1Ph and DS302 1Ph Gaskets kit p/n 9499370)<br/>(for pumps DS402 1Ph, DS402 3Ph, DS602 1Ph and DS602 3Ph Gaskets kit p/n 9499371)</li><li>Sight glass</li></ul> | <ul style="list-style-type: none"><li>Mechanical tools</li></ul> |



## 5 Technical Description

### Scheduled Maintenance Cards

#### **WARNING!**

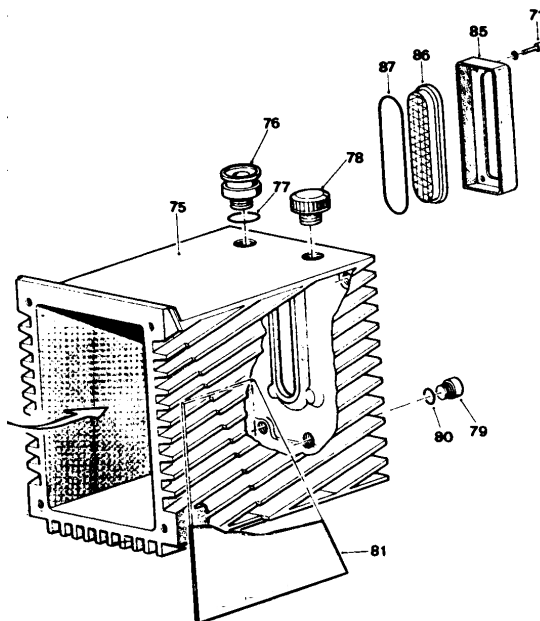


**Machine safety: disconnect all power lines. Hang the card "MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON" from the main breaker switch. When finished, remove the safety warning.**

#### **WARNING!**



**If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.**



1. Drain off the oil from the pump.

**NOTE**

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

---

2. Unscrew the securing screws (71) and disconnect the flange (85) from the casing.
3. Take out the OR gasket (87) and the sight glass (86).

**NOTE**

When reassembling, replace the gaskets that you have stripped down.

---

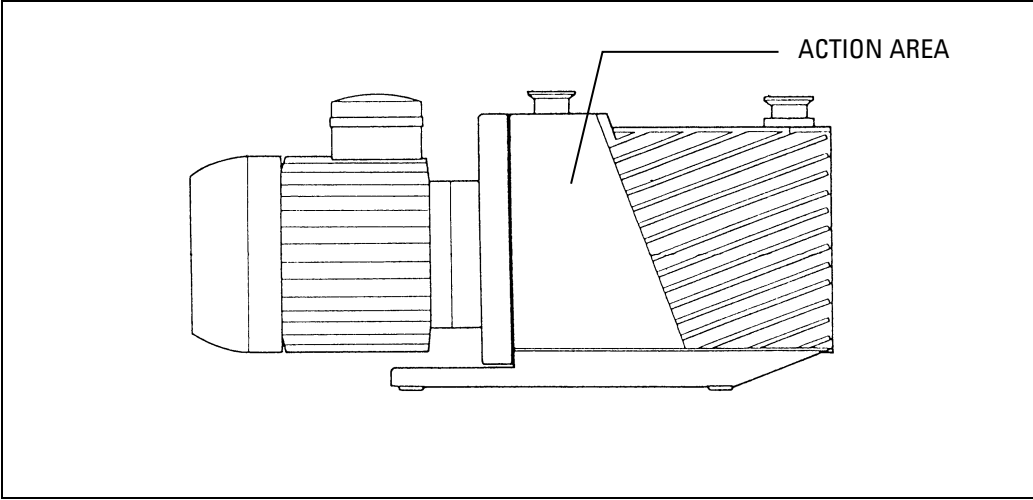
**CAUTION!**

When reassembly is complete, proceed to fill the pump with oil again following the instructions of Unscheduled Maintenance Card No. 102.

---

5    **Technical Description**  
Scheduled **Maintenance Cards**

| Unscheduled Maintenance |                                                                             | Card No.<br>104            |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 30 min                                                                | Machine status: <b>S</b> R |
| Type of action:         | REPLACEMENT                                                                 |                            |
| Item:                   | ANTI-SUCKBACK DEVICE SHUTTER                                                |                            |



|                     |                     |
|---------------------|---------------------|
| Materials required: | Equipment required: |
| ▪ Shutter           | ▪ Mechanical tools  |

**WARNING!**

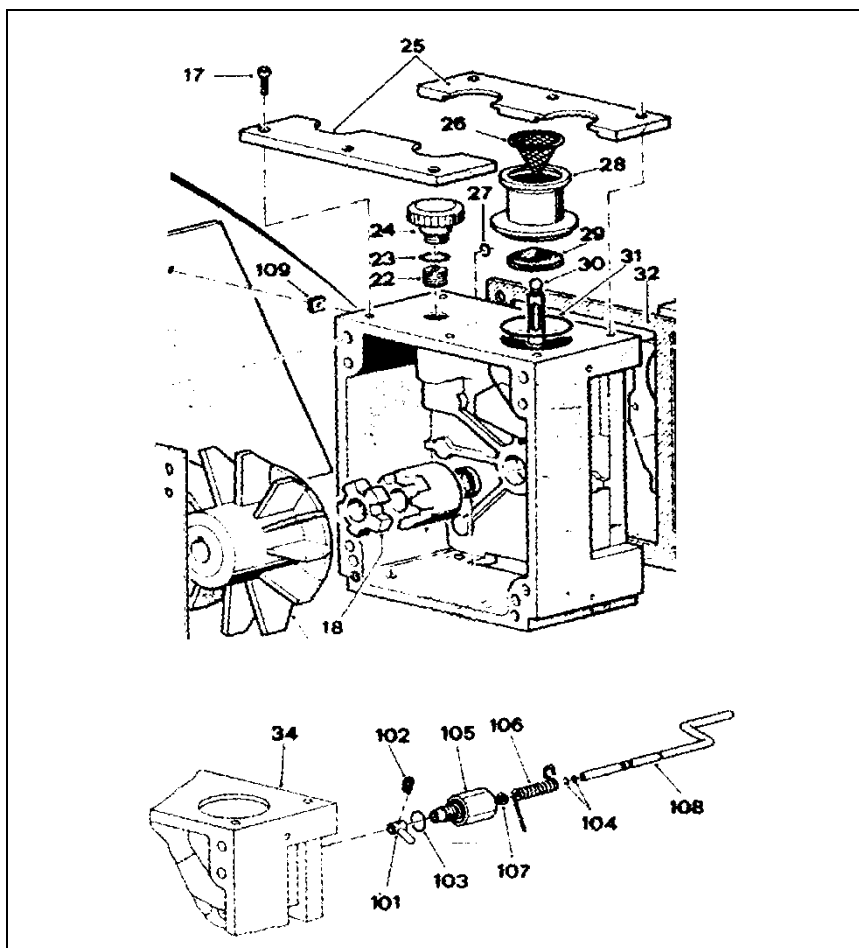


**Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.**

**WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.



## 5 Technical Description

### Scheduled Maintenance Cards

1. Disconnect the flange (28) from the inlet pipe line.
2. Remove the 6 securing screws (17) and remove the two half-plates (25) pressing the suction flange (28) into its seat.
3. Remove the flange (28) and take out the piston (30) with the shutter (29) of the anti-suckback device, disengage it by exerting force on arm (101). Separate the shutter (29) from the piston rod (30) simply by pulling.

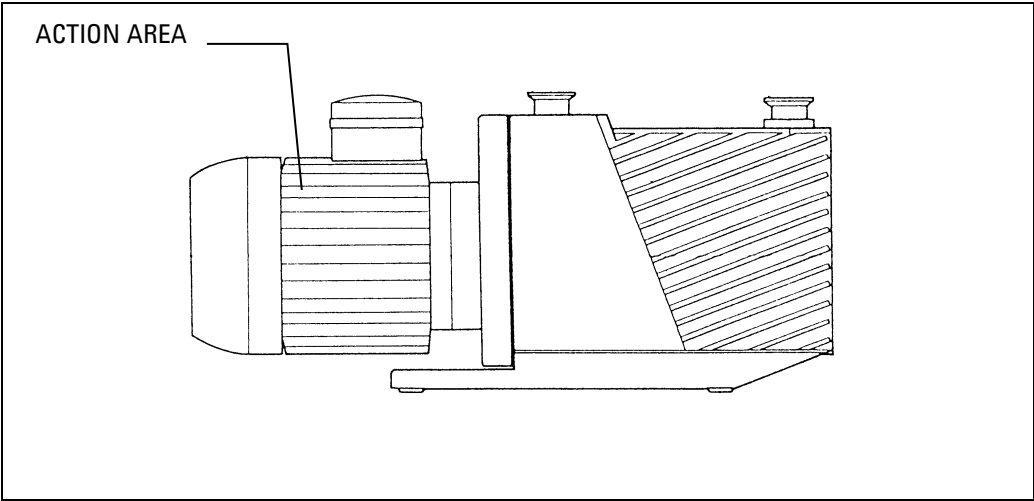
#### CAUTION!

Be careful not to scratch the sealing surface of the shutter (29).

---

4. Reassemble by proceeding in the inverse order.

| Unscheduled Maintenance |                                                                             | Card No.                   |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
|                         |                                                                             | 105                        |
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 30 min                                                                | Machine status: <b>S</b> R |
| Type of action:         | REPLACEMENT                                                                 |                            |
| Item:                   | ELECTRIC MOTOR AND JOINT                                                    |                            |



| Materials required: | Equipment required: |
|---------------------|---------------------|
| ▪ Electric motor    | ▪ Mechanical tools  |

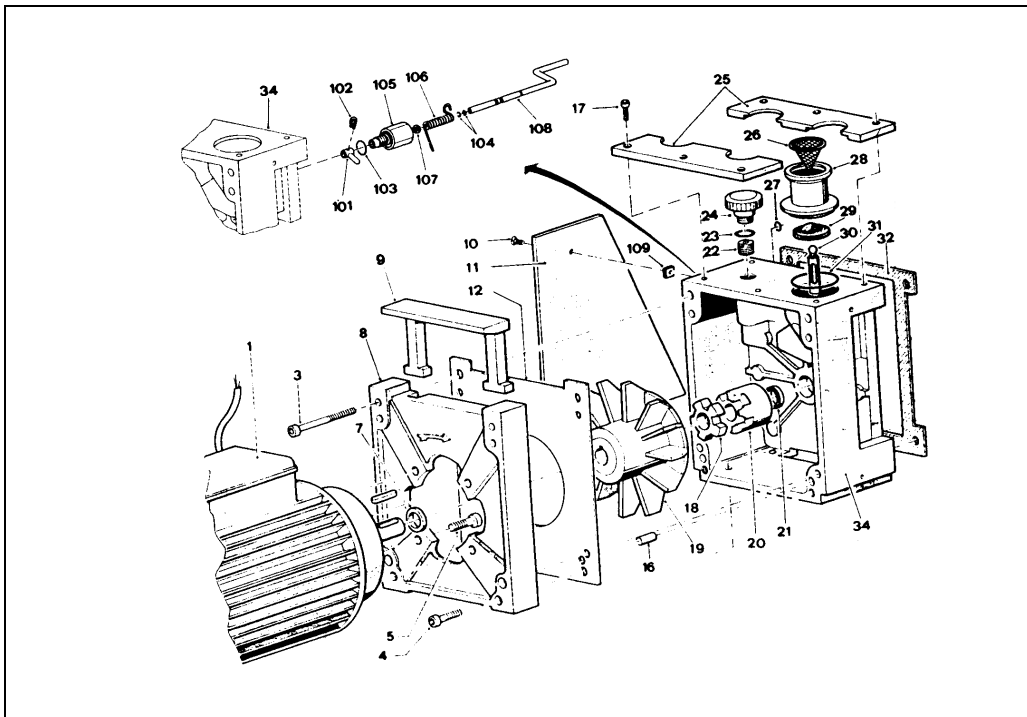
WARNING!

Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.

## Scheduled Maintenance Cards

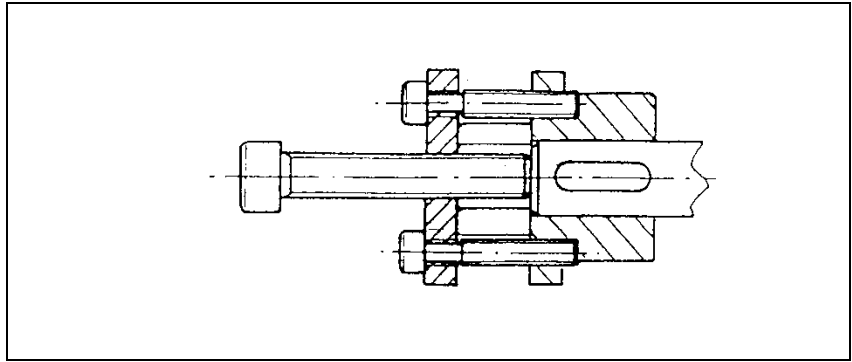


**If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.**



1. Disconnect the motor bearing plate (8) from the support (34) by removing the four screws (4).
2. Remove the half-joint/fan (19) from the support, together with the motor (1), the plate (8) and the plate (12) (for the DS402 1Ph, DS402 3Ph, DS602 1Ph and DS602 3Ph only).

3. Slacken the nut blocking the half-joint (19) to the motor shaft and take it out, using leverage between the plate (8) and the back of the joint.
4. Remove the four screws (5) and separate the plate (8) from the motor (1).
5. Reassemble by proceeding in the inverse order.

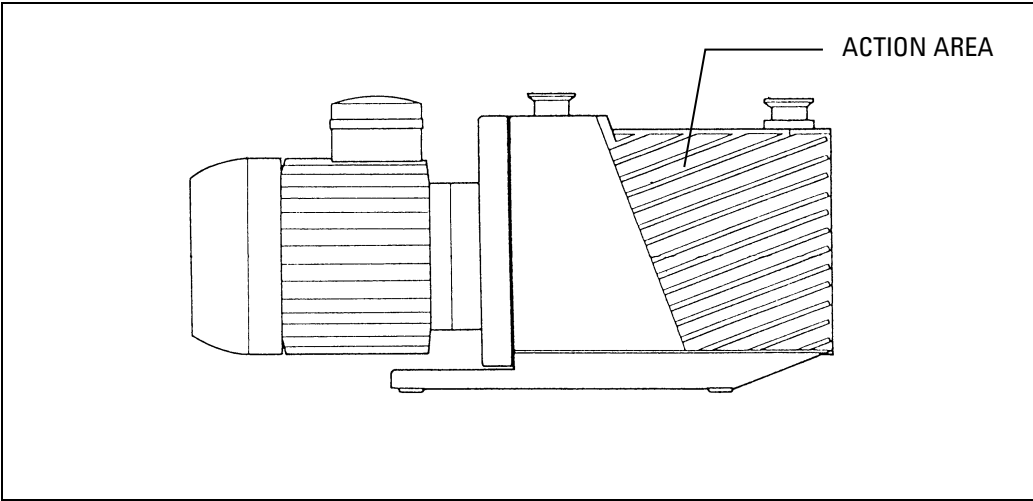


Extractor



**5    Technical Description**  
Scheduled **Maintenance Cards**

| Unscheduled Maintenance |                                                                             | Card No.<br>106            |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 1 hour                                                                | Machine status: <b>S</b> R |
| Type of action:         | DISASSEMBLY                                                                 |                            |
| Item:                   | DISCHARGE VALVES                                                            |                            |



|                        |                     |
|------------------------|---------------------|
| Materials required:    | Equipment required: |
| ▪ Valves (Gaskets kit) | ▪ Mechanical tools  |

**WARNING!**

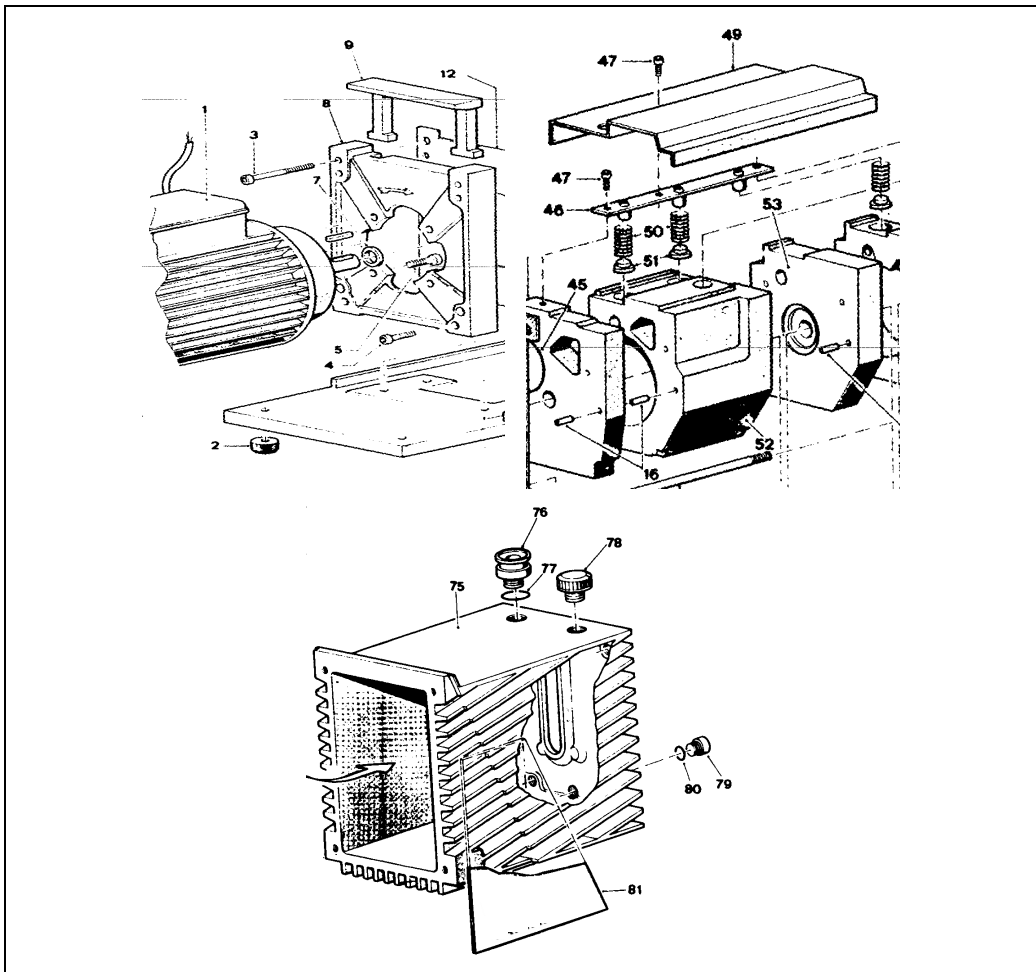


**Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.**

**WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.



## 5 Technical Description

### Scheduled Maintenance Cards

1. Drain off all the oil from the pump.

---

#### NOTE

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

---

2. Disconnect the discharge flange from its piping.
3. Remove the nuts (3) and remove the pump casing (75).
4. Remove the securing screw (47) holding the splash guard (49) and remove it.
5. Remove the two screws (47) on the valve press-bracket (46) and remove the bracket (46), the springs (50) and the shutters (51).
6. Reassemble by proceeding in the inverse order.

---

#### NOTE

When reassembling, replace all the gaskets stripped down.

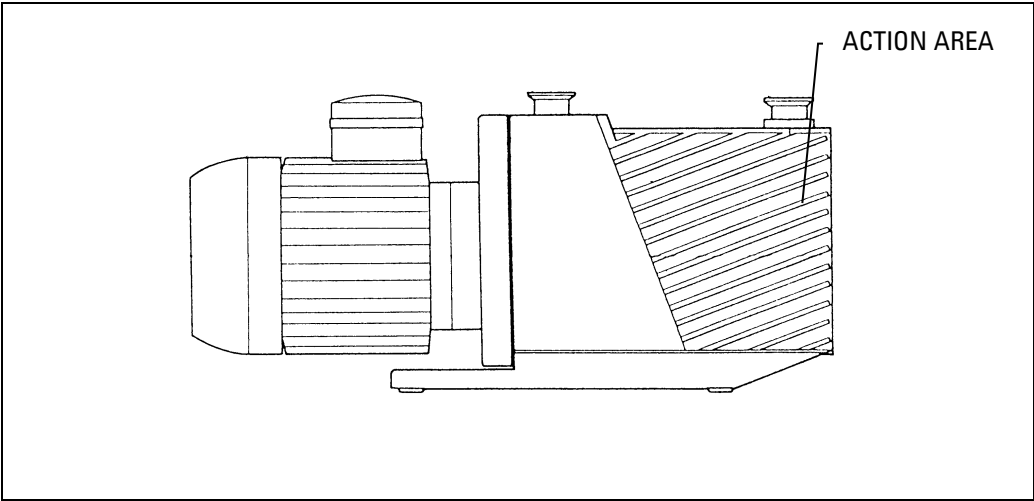
---

#### CAUTION!

When reassembly is complete, proceed to fill the pump with oil following the instructions of Unscheduled Maintenance Card No. 102.

---

| Unscheduled Maintenance |                                                                             | Card No.                   |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
|                         |                                                                             | 107                        |
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 1 hour                                                                | Machine status: <b>S</b> R |
| Type of action:         | DISASSEMBLY                                                                 |                            |
| Item:                   | OIL CIRCULATION PUMP 9499390                                                |                            |



| Materials required:                | Equipment required: |
|------------------------------------|---------------------|
| ▪ Oil circulating pump P/N 9499390 | ▪ Mechanical tools  |

WARNING!

Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.

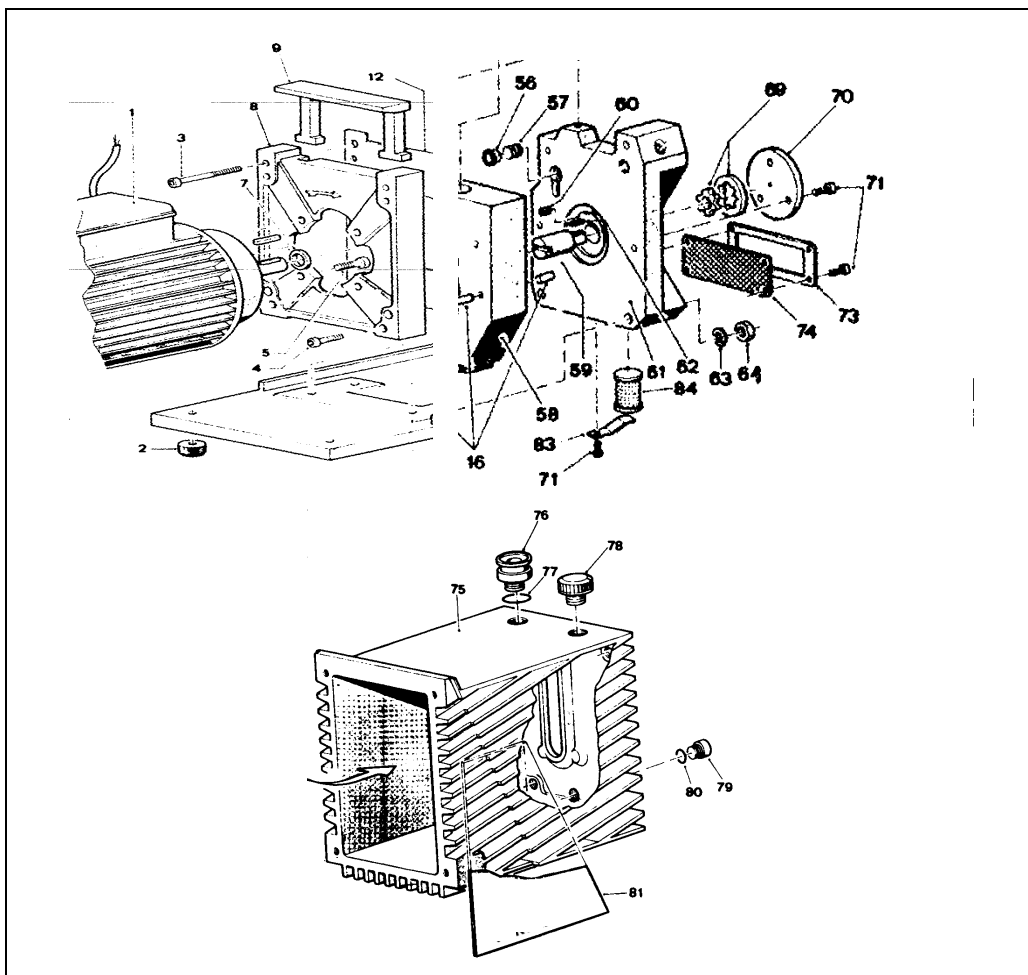
## 5 Technical Description

### Scheduled Maintenance Cards

#### **WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.



1. Drain off all the oil from the pump.

**NOTE**

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

---

2. Disconnect the discharge flange from its piping.
3. Remove the nuts (3) and remove the casing from the pump (75).
4. Remove the three screws (71) connecting the closing plate (70) to the side and take it out.
5. Remove the pump gear-wheels (69).

**NOTE**

A magnet may be used to remove the gear-wheels.

---

6. Reassemble by proceeding in the inverse order.

**NOTE**

When reassembling, replace all the gaskets you have stripped down.

---

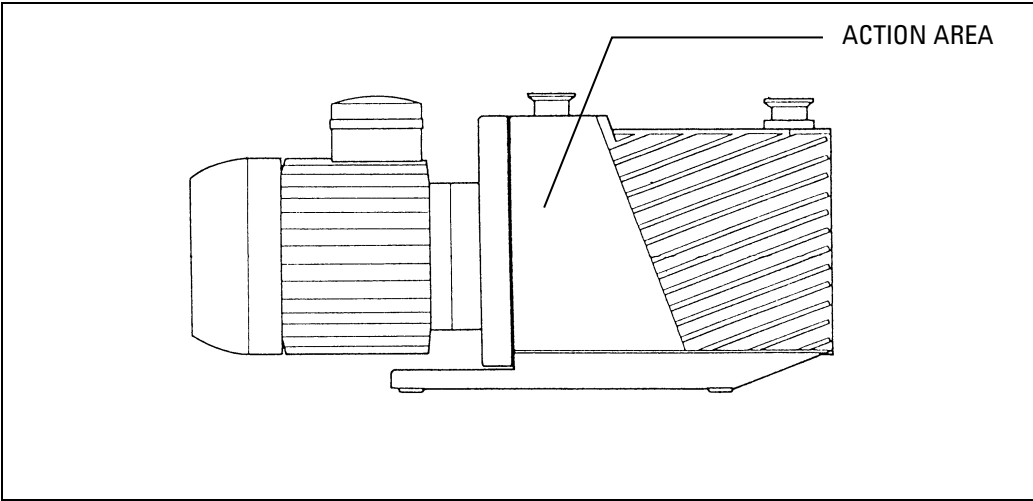
**CAUTION!**

When reassembly is complete, proceed to fill the pump with oil again following the instructions of Unscheduled Maintenance Card No. 102.

---

**5    Technical Description**  
Scheduled **Maintenance Cards**

| Unscheduled Maintenance |                                                                             | Card No.<br>108            |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 1 hour                                                                | Machine status: <b>S</b> R |
| Type of action:         | DISASSEMBLY                                                                 |                            |
| Item:                   | ANTI-SUCKBACK DEVICE                                                        |                            |



|                     |                     |
|---------------------|---------------------|
| Materials required: | Equipment required: |
| ▪ Gasket kit        | ▪ Mechanical tools  |

**WARNING!**

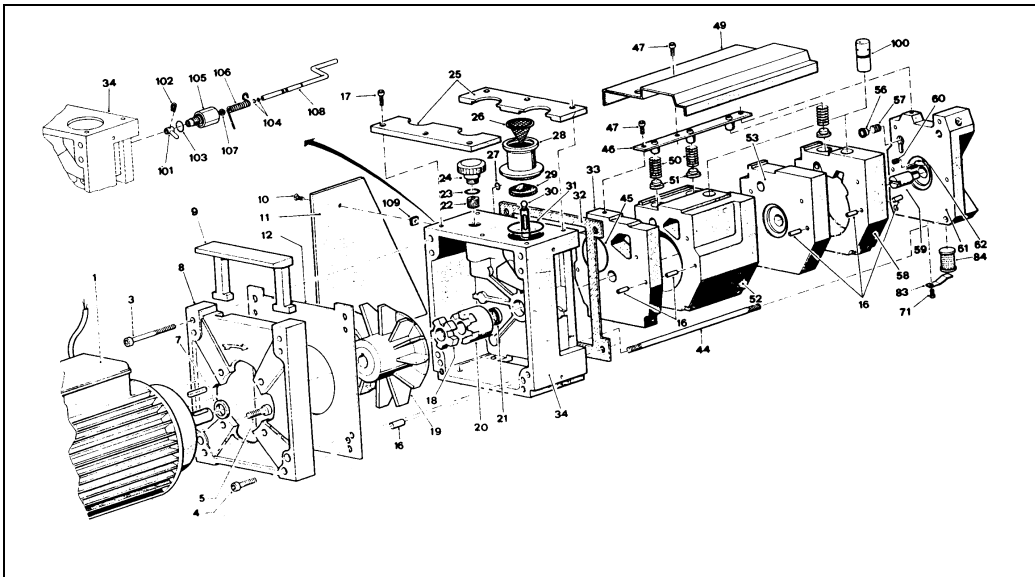


**Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.**

**WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.



1. Drain off all the oil from the pump.

**NOTE**

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

2. Disconnect the discharge flanges from their piping.
3. Remove the locking nuts (3) and remove the casing (75) from the pump.



## 5 Technical Description

### Scheduled Maintenance Cards

4. Remove the securing screw (47) and remove the splash guard (49).
5. Remove the shutter from the anti-suckback device.

#### NOTE

For more details on how to remove the anti-suckback device, see Unscheduled Maintenance Card No. 104.

---

6. Remove the screw (102), take out the arm (101) from the rod.
7. Release the spring (106) from the rod arm (108).
8. Pull the rod (108) out of the rotary feedthrough (105) sleeve.
9. Remove the two OR gaskets (104) from the rod (108) using a screwdriver, if necessary. Be careful not to bend the rod in doing so.
10. Remove the sleeve (105) and remove it from the support (34) together with its OR gasket (103).
11. Remove the hydraulic piston (100) from its seat in the body.
12. Reassemble by proceeding in the inverse order.

#### NOTE

When reassembling, replace all the gaskets stripped down.

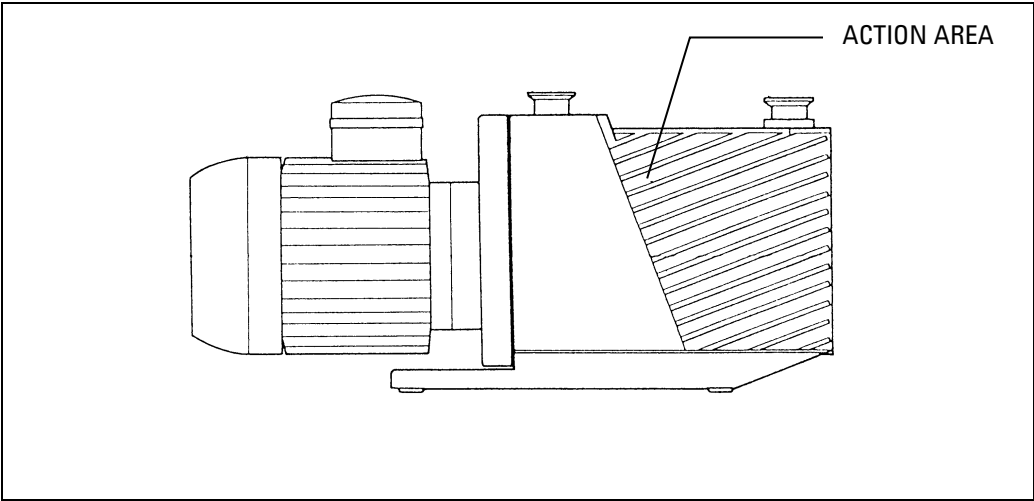
---

#### CAUTION!

When reassembly is complete, proceed to fill the pump with oil again following the instructions of Unscheduled Maintenance Card No. 102.

---

| Unscheduled Maintenance |                                                                             | Card No.<br>109            |
|-------------------------|-----------------------------------------------------------------------------|----------------------------|
| Pump type:              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                            |
| Frequency: WHEN DAMAGED | Time: 1 hour                                                                | Machine status: <b>S</b> R |
| Type of action:         | DISASSEMBLY                                                                 |                            |
| Item:                   | PUMP BODY                                                                   |                            |



| Materials required: | Equipment required: |
|---------------------|---------------------|
| ▪ Gasket kit        | ▪ Mechanical tools  |

WARNING!

Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.

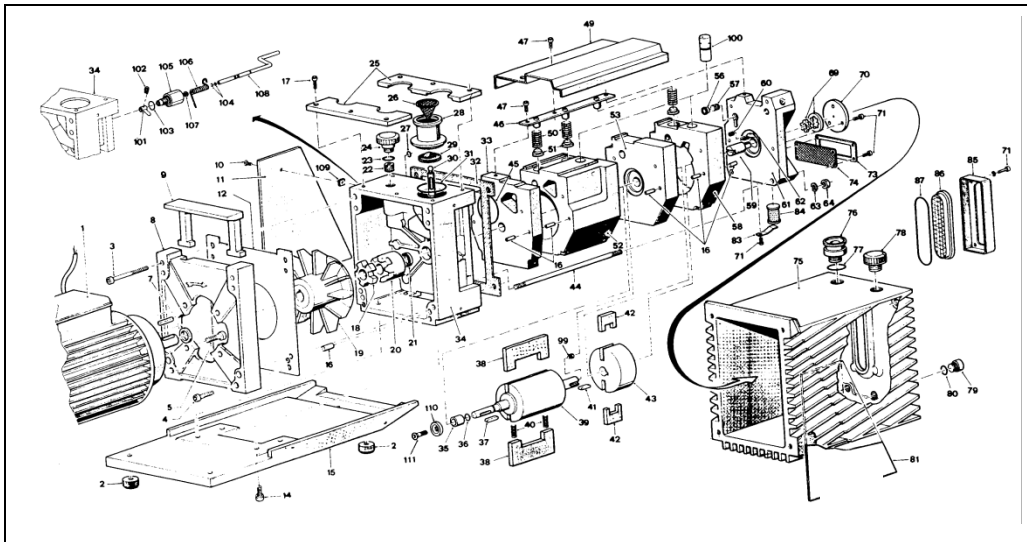
## 5 Technical Description

### Scheduled Maintenance Cards

#### WARNING!



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.



1. Drain off all the oil from the pump.

#### NOTE

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

2. Disconnect the discharge flanges from their piping.
3. Disconnect the electric motor (1) complete with half-joint.

**NOTE**

For a description of how to disassemble the electric motor, see the Unscheduled Maintenance Card No. 105.

---

4. On DS402 1Ph, DS402 3Ph, DS602 1Ph and DS602 3Ph: remove the screw (111) and remove the seal (110)). Slacken the nut locking the half-joint (20) to the shaft of the rotor (39) and take out the half-joint. Remove the tongue (37) with it as well.
5. Unscrew the securing screw (47) and remove the valve screen (49).
6. Slacken the four nuts (64) and take out the tie-rods (44) and the complete functional block (76), lifting it from the pump support (34) and take out the OR gaskets (27) and (45), from the gas ballast line and suction line respectively.

**NOTE**

Be careful not to cut the lip of the oil seal with the sharp edges of the shaft key seat.

---

7. Remove the screw (102), take out the arm (101) from the rod.

**NOTE**

When reassembling, replace all the gaskets stripped down.

---

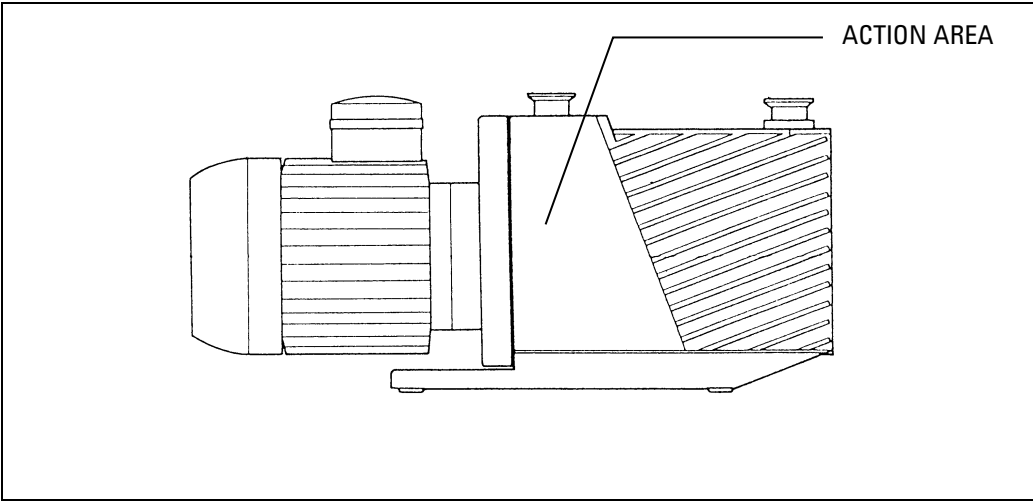
**CAUTION!**

When reassembly is complete, proceed to fill the pump with oil again following the instructions of Unscheduled Maintenance Card No. 102.

---

**5    Technical Description**  
Scheduled **Maintenance Cards**

| Unscheduled Maintenance        |                                                                             | Card No.<br>110                   |
|--------------------------------|-----------------------------------------------------------------------------|-----------------------------------|
| <b>Pump type:</b>              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                                   |
| <b>Frequency:</b> WHEN DAMAGED | <b>Time:</b> 1 hour and 15 min.                                             | <b>Machine status:</b> <b>S</b> R |
| <b>Type of action:</b>         | REPLACEMENT                                                                 |                                   |
| <b>Item:</b>                   | OIL SHAFT SEAL / ANTI-WEAR BUSHING                                          |                                   |



|                                                                                                                    |                                                                    |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Materials required:</b>                                                                                         | <b>Equipment required:</b>                                         |
| <ul style="list-style-type: none"><li>▪ Oil shaft seal</li><li>▪ Anti-wear bushing</li><li>▪ Gaskets kit</li></ul> | <ul style="list-style-type: none"><li>▪ Mechanical tools</li></ul> |

**WARNING!**

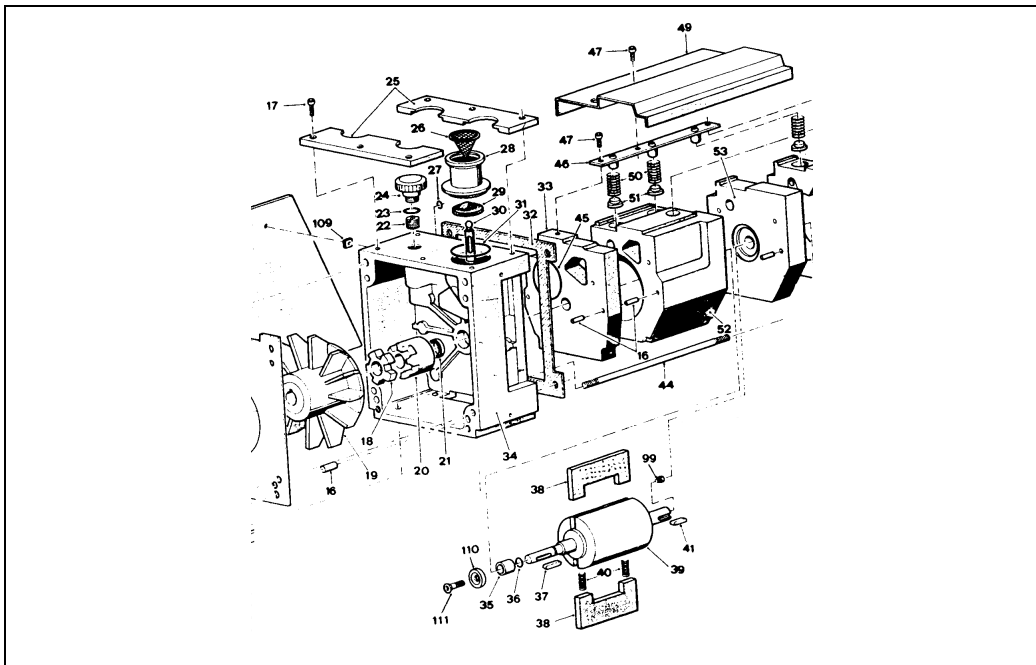


Machine safety: disconnect all power lines. Hang the card "MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON" from the main breaker switch. When finished, remove the safety warning.

**WARNING!**



If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.



## 5 Technical Description

### Scheduled Maintenance Cards

1. Drain off all the oil from the pump.

#### NOTE

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

---

2. Remove the worn out oil shaft seal (21) from the pump support (34) using a screwdriver for leverage and taking care not to scratch the seat.

#### NOTE

For a description of how to separate the functional block from the support, see the Unscheduled Maintenance Card No. 109.

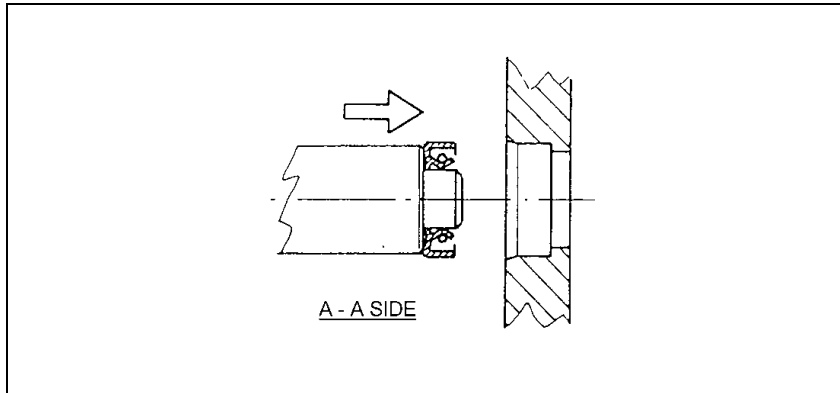
---

3. Remove the worn out oil shaft seal (21) from the pump support (34) using a screwdriver for leverage and taking care not to scratch the seat.
4. Take the anti-wear bushing (35) off the shaft and take out the OR Gasket (36).

#### NOTE

For assembly of the new oil shaft seal, a special tool is needed.

---



5. After applying silicone grease to the external diameter of the oil seal, engage it on the tool and push it into its seat.
6. Reassemble by proceeding in the inverse order.

**NOTE**

When reassembling, replace all the gaskets stripped down.

---

**CAUTION!**

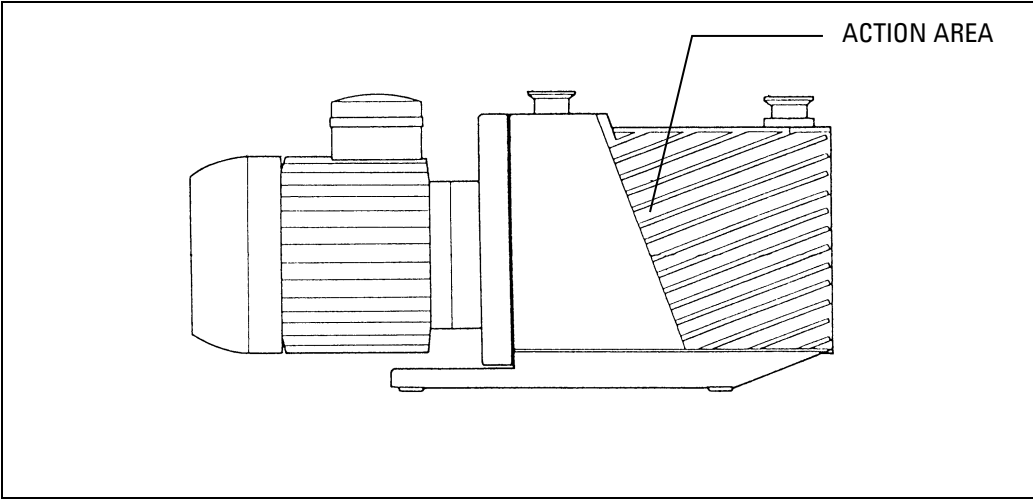
When reassembly is complete, proceed to fill the pump with oil again following the instructions of Unscheduled Maintenance Card No. 102.

---



**5    Technical Description**  
Scheduled **Maintenance Cards**

| Unscheduled Maintenance        |                                                                             | Card No.<br>111                   |
|--------------------------------|-----------------------------------------------------------------------------|-----------------------------------|
| <b>Pump type:</b>              | DS102 1Ph, DS202 1Ph, DS302 1Ph, DS402 1Ph, DS402 3Ph, DS602 1Ph, DS602 3Ph |                                   |
| <b>Frequency:</b> WHEN DAMAGED | <b>Time:</b> 2 hours                                                        | <b>Machine status:</b> <b>S</b> R |
| <b>Type of action:</b>         | DISASSEMBLY                                                                 |                                   |
| <b>Item:</b>                   | FUNCTIONAL BLOCK ROTORS                                                     |                                   |



| Materials required:                                                                                                                                                                                                                                                                     | Equipment required:                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>▪ Oil shaft seal ring</li><li>▪ Anti-wear bushing</li><li>▪ OR gaskets</li><li>▪ 1st stage vanes</li><li>▪ 2nd stage vanes</li><li>▪ Gas Ballast shutter</li></ul> <p>(the above items of material are included in the Major Maintenance kit)</p> | <ul style="list-style-type: none"><li>▪ Mechanical tools</li></ul> |

---

**WARNING!**



Machine safety: disconnect all power lines. Hang the card “MACHINE UNDERGOING MAINTENANCE - DO NOT POWER ON” from the main breaker switch. When finished, remove the safety warning.

---

---

**WARNING!**

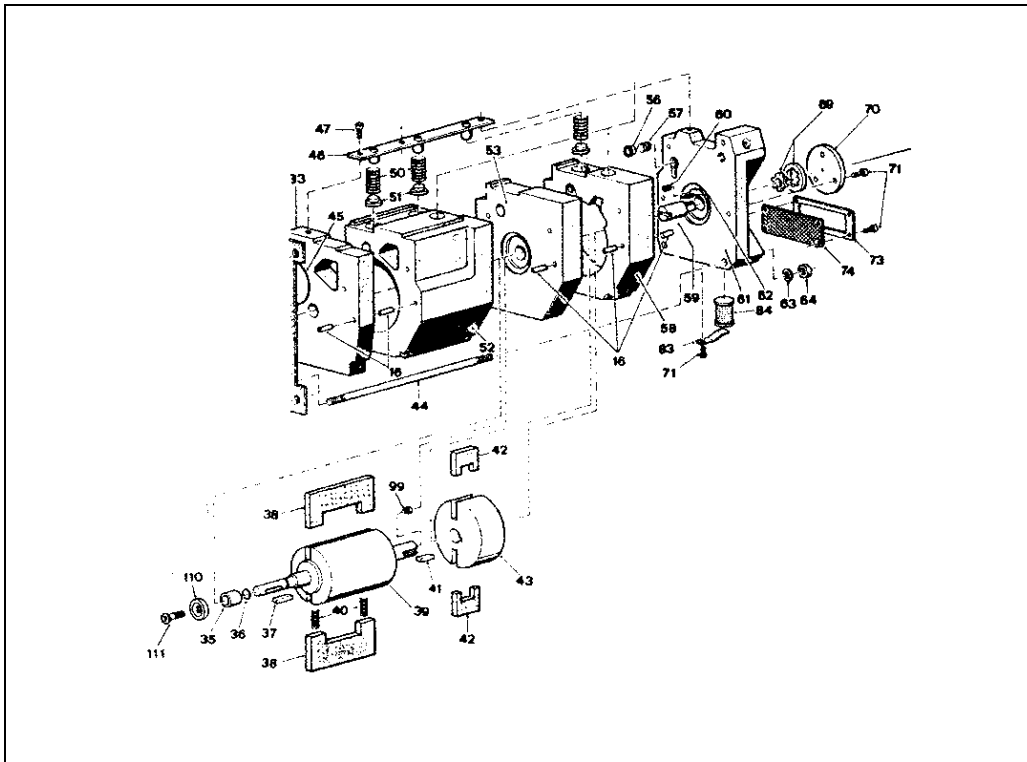


If you have to perform maintenance on the pump after a considerable time in operation, leave it to cool as temperature of the outer surface may be in excess of 60 °C.

---

## 5 Technical Description

### Scheduled Maintenance Cards



1. Drain off all the oil from the pump.

#### NOTE

For a description of how to drain off the oil, see the Unscheduled Maintenance Card No. 102.

2. Detach the entire functional block from the pump support.

#### NOTE

For a description of how to separate the functional block from the support, see the Unscheduled Maintenance Card No. 109.

3. Disassemble the discharge valves.

**NOTE**

For a description of how to disassemble the discharge valves, see the Unscheduled Maintenance Card No. 106.

---

4. Detach the end plate (61) from the rest of the pump body, exerting leverage if necessary on the reference notches between the parts. This will give you access to the retaining valve (56) and associated spring (57) at the gas ballast inlet.
5. Disconnect the 2nd stage stator (58), remove the two vanes (42), take the 2nd stage rotor (43) off the shaft and take out the key (41).
6. Remove the partition wall (53) from between the 1st and 2nd stages.
7. Detach the 1st stage stator (52) from the inner side (33), pressing back on the springs (40) of the vanes (38) to avoid the vanes from touching against the sharp corners of the body and becoming damaged.
8. Take out the 1st stage rotor (39), the vanes (38) and the springs (40).
9. Reassemble by proceeding in the inverse order.

**NOTE**

When reassembling, replace all the gaskets stripped down.

---

**CAUTION!**

After reassembly is complete, proceed to fill the pump again with oil following the instructions of Unscheduled Maintenance Card No. 102.

---

## Troubleshooting and Corrective Action

### NOTE

If the pump breaks down or does not work well, first check thoroughly that all the standards rules listed in the sections relevant installation and operation are currently observed.

If you cannot get sufficiently low pressure, the problem may lie with the system connected to the pump, i.e. the vacuum chamber and inlet line.

Even when the design sizes are correct, the problems listed below are still fairly frequent:

- a) faulty seals: check that the seals are fitted, are in good working order and are clean; check all joints, taps, sight glasses, weld joints, etc.
- b) unsuitable materials: avoid the use of rubber, plastic or even metal parts that are porous or permeable.
- c) volatile substances: as far as possible, eliminate humidity, grease, dirt, rust, solvents (even small traces).

**Tab. 12**

|   | SYMPTOM                                         | PROBABLE CAUSE                           | SUGGESTED REMEDY                                                                                                                                         | CARD REF. |
|---|-------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1 | The pump does not start (or stops immediately). | A) The motor-pump coupling is defective. | Check state of the elastic joint and replace the parts as needed.                                                                                        | 105       |
|   |                                                 | B) Viscosity of the oil has increased.   | Examine the oil and change if necessary (find out why it has deteriorated).                                                                              | 102       |
|   |                                                 | C) Excessive friction inside pump.       | Test the resistance to rotation, by turning the joint by hand. If the rotors are blocked, disassemble the pump body and check state of the parts inside. | 111       |

| SYMPTOM |                                          | PROBABLE CAUSE                                                       | SUGGESTED REMEDY                                                                          | CARD REF.                                  |
|---------|------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|
|         |                                          | D) Electrical protections not set.                                   | Adjust the protections.                                                                   |                                            |
|         |                                          | E) Oil too viscous because of low room temperature.                  | Heat the oil up to a temperature of 20 °C or use an oil recommended for low temperatures. | 102                                        |
|         |                                          | F) Direction of rotation inverted (only 3-phase motors).             | Change over two phase wires of the electric power supply.                                 |                                            |
| 2       | The pump stops (stalls).                 | A) Seizure of some internal parts.                                   | Proceed as at 1-C.                                                                        |                                            |
|         |                                          | B) Oil very dirty.                                                   | Proceed as at 1-B.                                                                        |                                            |
| 3       | The pump overheats.                      | A) Too much friction inside.                                         | Proceed as at 1-C.                                                                        |                                            |
|         |                                          | B) Oil ducts clogged inside.                                         | Try changing the oil, otherwise disassemble the pump and clean it well.                   | 102<br>111                                 |
| 4       | Too much oil drained off by the exhaust. | A) Gas Ballast device broken (always open).                          | Check the device to ensure the gas ballast is properly adjusted.                          | 111                                        |
|         |                                          | B) Heavy duty work (high suction pressures or very frequent cycles). | Check condition of the oil trap filter.                                                   |                                            |
|         |                                          | C) Leak in system.                                                   | Check the system (see "Note").                                                            |                                            |
| 5       | The pump is loosing oil.                 | A) Seal on shaft defective.                                          | Replace the oil seal and the anti-wear bushing.                                           | 110                                        |
|         |                                          | B) Other gaskets defective.                                          | Identify the areas where seeping occurs and check the associated gaskets.                 | Refer to the card related to involved item |
|         |                                          | C) Oil level window broken                                           | Replace the oil level window.                                                             |                                            |

## 5 Technical Description

### Troubleshooting and Corrective Action

|   | SYMPTOM                                                     | PROBABLE CAUSE                                             | SUGGESTED REMEDY                                                                                                          | CARD REF. |
|---|-------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------|
| 6 | The pump is very noisy even at steady operating conditions. | A) Exhaust valves damaged.                                 | Inspect the valves.                                                                                                       | 106       |
|   |                                                             | B) High suction pressure, or gas ballast device is open.   | Check the system and the gas ballast valve (see items 4-A and 4-B).                                                       |           |
|   |                                                             | C) Elastic star of joint is damaged.                       | Check and replace if necessary.                                                                                           |           |
|   |                                                             | D) A screw of the joint has become loose.                  | Lock the screw after application of Loctite 222 (for slack threads).                                                      |           |
| 7 | When the pump is stopped, the inlet pressure rises rapidly. | A) The anti-suckback device is defective.                  | Disassemble and check the anti-suckback device.                                                                           | 108       |
|   |                                                             | B) The cause of the problem is not with the pump.          | See "Note".                                                                                                               |           |
| 8 | The pump does not produce a sufficient vacuum.              | A) Not enough oil, or oil contaminated.                    | Check level and appearance of the oil. Open the Gas Ballast for half an hour at least; change the oil whenever necessary. | 01<br>102 |
|   |                                                             | B) Oil ducts clogged inside.                               | Proceed as at 3-B.                                                                                                        |           |
|   |                                                             | C) Direction of rotation inverted.                         | Proceed as at 3-C.                                                                                                        |           |
|   |                                                             | D) OR gaskets communicating with the inlet line defective. | Replace the gaskets.                                                                                                      | 109       |
|   |                                                             | E) Exhaust valves damaged or out of place.                 | Proceed as at 6-A.                                                                                                        |           |

| SYMPTOM | PROBABLE CAUSE                                                                                                                     | SUGGESTED REMEDY                                                                    | CARD REF.  |
|---------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------|
|         | F) Insufficient oil pressure in the lubricating circuit, preventing the anti-suckback device from opening.                         | Inspect the gears of the oil pump and check that it is working.                     | 107        |
|         | G) Internal moving parts worn.                                                                                                     | Disassemble the pump, and replace any worn parts.                                   | 111        |
|         | H) Pump working with gas ballast device open.                                                                                      | Close the gas ballast device.                                                       |            |
|         | I) Anti-suckback valve fails to open.                                                                                              | Disassemble the suction flange and check the device.                                | 104<br>108 |
|         | J) Parts of the functional block not in perfect contact because of deposits of dirt or failure to put a plug on the service holes. | Disassemble and check condition of the part surfaces and that plugs have been used. | 111        |
|         | K) The cause of the problem is not with the pump.                                                                                  | See "Note".                                                                         |            |



## Spare Parts List

**Tab. 13**

| OTHER SPARE PARTS          | Q.TY      |           |           |                        |                        | P/N         |
|----------------------------|-----------|-----------|-----------|------------------------|------------------------|-------------|
|                            | DS102 1Ph | DS202 1Ph | DS302 1Ph | DS402 1Ph<br>DS402 3Ph | DS602 1Ph<br>DS602 3Ph |             |
| MINOR MAINTENANCE KIT (*)  | X         | X         | X         |                        |                        | 9499370     |
| MINOR MAINTENANCE KIT (*)  |           |           |           | X                      | X                      | 9499371     |
| MAJOR MAINTENANCE KIT (**) | X         |           |           |                        |                        | 9499380     |
| MAJOR MAINTENANCE KIT (**) |           | X         | X         |                        |                        | 9499381     |
| MAJOR MAINTENANCE KIT (**) |           |           |           | X                      | X                      | 9499382     |
| COUPLING KIT (***)         | X         | X         | X         |                        |                        | 9499370M002 |
| COUPLING KIT (***)         |           |           |           | X                      | X                      | 9499371M002 |

(\*) Contains all the valves, O-rings and seals to refurbish the pump to vacuum integrity.

(\*\*) Includes all the items of the minor maintenance kit plus the vanes.

(\*\*\*)Includes items 18, 19 and 20 shown in the exploded view of page 146.

**Tab. 14**

| ACCESSORY ITEM ON EXHAUST   | Q.TY      |           |           |                        |                        | P/N     |
|-----------------------------|-----------|-----------|-----------|------------------------|------------------------|---------|
|                             | DS102 1Ph | DS202 1Ph | DS302 1Ph | DS402 1Ph<br>DS402 3Ph | DS602 1Ph<br>DS602 3Ph |         |
| OIL MIST ELIMINATOR KF25    | X         | X         | X         | X                      | X                      | 9499395 |
| OIL MIST CARTRIDGE (Q.ty 2) | X         | X         | X         | X                      | X                      | 9499394 |

**Tab. 15**

| <b>RECOMMENDED OIL</b>                          | <b>P/N</b> | <b>Q.TY</b> |
|-------------------------------------------------|------------|-------------|
| Agilent Rotary Vane Fluid<br>DS19 Type (Europe) | 9499390    | 1 liter     |
| Agilent Rotary Vane Fluid<br>DS19 Type (USA)    | K7516301   | 1 liter     |
| Agilent Rotary Vane Fluid<br>DS19 Type (USA)    | K7516302   | 1 gallon    |

## 5 Technical Description

### Spare Parts List



## ***Vacuum Products Division***

*Dear Customer,*

*Thank you for purchasing an Agilent vacuum product. At Agilent Vacuum Products Division we make every effort to ensure that you will be satisfied with the product and/or service you have purchased.*

*As part of our Continuous Improvement effort, we ask that you report to us any problem you may have had with the purchase or operation of our products. On the back side you find a Corrective Action request form that you may fill out in the first part and return to us.*

*This form is intended to supplement normal lines of communications and to resolve problems that existing systems are not addressing in an adequate or timely manner.*

*Upon receipt of your Corrective Action Request we will determine the Root Cause of the problem and take the necessary actions to eliminate it. You will be contacted by one of our employees who will review the problem with you and update you, with the second part of the same form, on our actions.*

*Your business is very important to us. Please, take the time and let us know how we can improve.*

*Sincerely,*

A handwritten signature in black ink, appearing to read "Giampaolo LEVI".

**Giampaolo LEVI**

*Vice President and General Manager  
Agilent Vacuum Products Division*

**Note:** Fax or mail the Customer Request for Action (see backside page) to Agilent Vacuum Products Division (Torino) – Quality Assurance or to your nearest Agilent representative for onward transmission to the same address.

## CUSTOMER REQUEST FOR CORRECTIVE / PREVENTIVE / IMPROVEMENT ACTION

TO: AGILENT VACUUM PRODUCTS DIVISION TORINO – QUALITY ASSURANCE FAX

N°: XXXX-011-9979350

ADDRESS: AGILENT TECHNOLOGIES ITALIA S.p.A. – Vacuum Products Division –

Via F.lli Varian, 54 – 10040 Leini (TO) – Italy

E-MAIL: [vpd-qualityassurance\\_pdl-ext@agilent.com](mailto:vpd-qualityassurance_pdl-ext@agilent.com)

| NAME                                                                                                         | COMPANY | FUNCTION |
|--------------------------------------------------------------------------------------------------------------|---------|----------|
| ADDRESS:                                                                                                     |         |          |
| TEL. N° : FAX N° :                                                                                           |         |          |
| E-MAIL:                                                                                                      |         |          |
| PROBLEM / SUGGESTION :                                                                                       |         |          |
| REFERENCE INFORMATION (model n°, serial n°, ordering information, time to failure after installation, etc.): |         |          |
| DATE                                                                                                         |         |          |
| CORRECTIVE ACTION PLAN / ACTUATION<br>(by AGILENT VPD)                                                       |         | LOG N°   |

XXX = Code for dialing Italy from your country (es. 01139 from USA; 00139 from Japan, etc.)

## **Vacuum Products Division Instructions for returning products**

Dear Customer,

Please follow these instructions whenever one of our products needs to be returned.

Complete the attached **Request for Return form** and send it to Agilent Technologies (see below), taking particular care to include the completed **Health and Safety** declaration Section. No work can be started on your unit until we receive a completed copy of this form.

After evaluating the information, Agilent Technologies will provide you with a **Return Authorization (RA) number** via email or fax, as requested. Note: Depending on the type of return, a Purchase Order may be required at the time **the Request for Return is submitted**. We will quote any necessary services (evaluation, repair, special cleaning, eg).

### **Product preparation**

- Remove all accessories from the core product (e.g. inlet screens, vent valves).
- Prior to shipment and if applicable for your product, drain any oils or other liquids, purge or flush all gasses, and wipe off any excess residue.
- If ordering an Advance Exchange product, please use the packaging from the Advance Exchange to return the defective product.
- Seal the product in a plastic bag, and package product carefully to avoid damage in transit. You are responsible for loss or damage in transit.
- Include a copy of the Health and Safety Declaration in the shipping documentation on the outside of the shipping box of your returning product.
- Clearly label package with RA number. Using the shipping label provided will ensure the proper address and RA number are on the package. Packages shipped to Agilent without a RA clearly written on the outside cannot be accepted and will be returned.
- Return only products for which the RA was issued.

### **Shipping**

- Ship to the location specified on the printable label, which will be sent, along with the RA number, as soon as we have received all of the required information. Customer is responsible for freight charges on returning product.
- Return shipments must comply with all applicable Shipping Regulations (IATA, DOT, ADR, etc.) and carrier requirements.

**RETURN THE COMPLETED REQUEST FOR RETURN FORM TO YOUR NEAREST LOCATION:**

#### **EUROPE:**

Fax: 00 39 011 9979 330  
Fax Free: 00 800 345 345 00  
Toll Free: 00 800 234 234 00

#### **NORTH AMERICA:**

Fax: 1 781 860 9252  
Toll Free: 800 882 7426, Option 3

#### **PACIFIC RIM:**

Please visit our website for individual office information

[vpt-customer care@agilent.com](mailto:vpt-customer care@agilent.com)

[vpl-ra@agilent.com](mailto:vpl-ra@agilent.com)

<http://www.agilent.com>

**TERMS AND CONDITIONS**

**Please read the terms and conditions below as they apply to all returns and are in addition to the Agilent Technologies Vacuum Product Division – Products and Services Terms of Sale.**

- Unless otherwise pre-negotiated, customer is responsible for the freight charges for the returning product. Return shipments must comply with all applicable **Shipping Regulations** (IATA, DOT, etc.) and carrier requirements.
- Agilent Technologies is not responsible for returning customer provided packaging or containers.
- Customers receiving an Advance Exchange product agree to return the defective, rebuildable part to Agilent Technologies **within 15 business days**. Failure to do so, or returning a non-rebuildable part (crashed), will result in an invoice for the non-returned/non-rebuildable part.
- Returns for credit toward the purchase of new or refurbished Products are subject to prior Agilent approval and may incur a restocking fee. Please reference the original purchase order number.
- Units returned for evaluation will be evaluated, and a quote for repair will be issued. If you choose to have the unit repaired, the cost of the evaluation will be deducted from the final repair pricing. A Purchase Order for the final repair price should be issued within 3 weeks of quotation date. Units without a Purchase Order for repair will be returned to the customer, and the evaluation fee will be invoiced.
- Products returned that have not been drained from oil will be disposed.
- A Special Cleaning fee will apply to all exposed products.
- If requesting a calibration service, units must be functionally capable of being calibrated.



| Customer information |  |               |
|----------------------|--|---------------|
| Company :            |  | Contact Name: |
| Address:             |  | Tel: Fax:     |
|                      |  | Email:        |

| Equipment           |                |                                                    |                               |
|---------------------|----------------|----------------------------------------------------|-------------------------------|
| Product description | Agilent PartNo | Agilent Serial No                                  | Original Purchasing Reference |
| Failure description |                | Type of process (for which the equipment was used) |                               |

| Type of return                                                                                                                                                                                                                                       |                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <input type="checkbox"/> Non Billable <input type="checkbox"/> Billable                                                                                                                                                                              | ➔ New PO # (hard copy must be submitted with this form): _____ |
| <input type="checkbox"/> Exchange <input type="checkbox"/> Repair <input type="checkbox"/> Upgrade <input type="checkbox"/> Demo <input type="checkbox"/> Calibration <input type="checkbox"/> Evaluation <input type="checkbox"/> Return for Credit |                                                                |

| Health and safety                                                                                                          |                                                          | Substances (please refer to MSDS forms)                                                                                                                                           |               |                 |            |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|------------|
| The product has been exposed to the following substances:<br>(by selecting 'YES' you MUST complete the table to the right) |                                                          | * Agilent will not accept delivery of any product that is exposed to radioactive, biological, explosive substances or dioxins, PCB's without written evidence of decontamination. |               |                 |            |
|                                                                                                                            |                                                          | Trade name                                                                                                                                                                        | Chemical name | Chemical Symbol | CAS Number |
| Toxic                                                                                                                      | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Harmful                                                                                                                    | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Corrosive                                                                                                                  | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Reactive                                                                                                                   | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Flammable                                                                                                                  | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Explosive (*)                                                                                                              | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Radioactive (*)                                                                                                            | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Biological (*)                                                                                                             | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Oxidizing                                                                                                                  | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Sensitizer                                                                                                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |
| Other dangerous substances                                                                                                 | <input type="checkbox"/> YES <input type="checkbox"/> NO |                                                                                                                                                                                   |               |                 |            |

| Goods preparation                                                                   |                                                                      |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| If you have replied YES to one of the above questions. Has the product been purged? | <input type="checkbox"/> YES <input type="checkbox"/> NO             |
| If yes, which cleaning agent/method:                                                |                                                                      |
| Has the product been drained from oil?                                              | <input type="checkbox"/> YES <input type="checkbox"/> NOT APPLICABLE |
| I confirm to place this declaration on the outside of the shipping box.             |                                                                      |

| I declare that the above information is true and complete to the best of my knowledge and belief.<br>I understand and agree to the terms and conditions on page 2 of this document.                                                                                                                                                                                                                                                 |                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Name:                                                                                                                                                                                                                                                                                                                                                                                                                               | Authorized Signature: |
| Position:                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |
| Date:                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |
| <b>NOTE:</b> If a product is received at Agilent which is contaminated with a toxic or hazardous material that was not disclosed, <b>the customer will be held responsible</b> for all costs incurred to ensure the safe handling of the product, and is liable for any harm or injury to Agilent employees as well as to any third party occurring as a result of exposure to toxic or hazardous materials present in the product. |                       |



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